The 11th International Workshop on Compositional Data Analysis

Book of Abstracts

1-5 June 2026, Coimbra, Portugal

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Welcome to CoDaWork 2026

The 11th International Workshop on Compositional Data Analysis

CoDaWork2026, the 11th International Workshop on Compositional Data Analysis, offers an open forum for discussing the latest advances in the methodology and applications of compositional data analysis and related methods in different fields. The primary goal of the workshop is to present the state of the art and to identify relevant lines of future methodological research as well as areas of applications; bringing together researchers from varied disciplines, academic mathematicians and statisticians, applied data analysts and modellers, research students, as well as those with a general interest in the field, to share their ideas, experiences, and developments. The Scientific Committee has aimed to craft a balanced and appealing programme covering the diverse interests of the participants. The Organising Committee has taken care of all the details, aiming to provide an inspiring and fruitful environment for discussion and to foster networking and collaboration. The current edition brings on a wealth of methodological contributions, ranging from those related to the basic foundations to those presenting novel developments in areas such as regression analysis, clustering, zero treatment, or dealing with high-dimensional compositions; along with innovative applications comprising the geological and environmental sciences, human health, molecular biology, and socio-economic applications, amongst others. Along with the traditional introductory course to compositional data analysis held on the first day, this edition includes abstract of 55 contributions. The abstracts of the presentations and posters are included in this book in alphabetical order. Finally, we acknowledge the effort and work of all those involved in organising CoDaWork2026, as well as the support of the CoDa-Association, sponsors, and government bodies.

We wish you a productive and stimulating CoDaWork 2026 and a great stay in Coimbra.

Yours sincerely,

Teresa Albuquerque, Chair of the Organising Committee.
Juan José Egozcue, Chair of the Scientific Committee.

Organising Committee

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Instructors of the Introductory Compositional Data Analysis Course

María Isabel Ortego and Iván Galván-Femenía

CoDaWork 2026 schedule

COLOR CODE

Methods	**Microbiology**	**Earth-sciences**
Health	**Social sciences**	**Invited talks**
Poster session	Breaks	Activities

	Tuesday	**Wednesday**	**Thursday**	**Friday**
9:00–9:20		Kanjiradan et al.	Erb et al.	Mondon et al.
9:20–9:40	Registration	McKinley et al.	Eilers	Higgins
9:40–10:00	Opening ceremony	Miatke et al.	Tolosana-D	Dos Santos et al.
10:00–10:20	Ascari et al.	Morimoto	Pavlů	Araiz et al.
10:20–10:40	Comas-Cufí et al.	Narayana et al.	Saperas-R et al.	Galván-F et al.
10:40–11:00	Czolková et al.	Prinelli	Sebatjane et al.	Creus-Marti et al.
11:00–11:50	coffee break	coffee break	coffee break	coffee break
11:50–12:10	Silva-Solar et al.	Ferrer-R et al.	Ševčíková et al.	Awards and
12:10–12:30	Calle et al.	Vega-B et al.	Stanford et al.	Closing ceremony
12:30–12:50	Te Beest et al.	Yoshida et al.	Tang N et al.	
12:50–14:30	lunch	lunch	lunch	
14:30–15:30	Paul-Gauthier		Genius	
15:30–15:50	Gijtenbeek et al.		Fačevicová et al.	
15:50–16:10	Engle et al.		Faugueras	
16:10–16:30	Valls		Nakamura	
16:30–16:50	Wehner		Oviedo-C et al.	
16:50–17:10	Rezaei et al.		Škorňa et al.	
17:10–17:30			Ricciotti et al.	
17:30–18:00	coffee break		coffee break	
18:00–19:00	Poster session	Excursion	CoDa-Assoc. G. Assembly	

CoDaWork 2026 general schedule

Monday

09:00 – 10:00	Registration
10:00 – 11:00	Introductory CoDa-Course. Instructors: Ortego MI, Galván-Femenía I
11:00 – 11:50	**Coffee-break**
11:50 – 12:50	Introductory CoDa-Course
12:50 – 14:30	**Lunch**
14:30 – 16:30	Introductory CoDa-Course
16:30 – 17:00	**Coffee-break**
17:00 – 19:00	Introductory CoDa-Course
19:00 – 21:30	Reception – Ice breaker

Tuesday

09:00 – 09:40	**Registration**
09:40 – 10:00	**Opening ceremony**
Methods I	**chair: Fačevicová K**
10:00 – 10:20	Ascari R, Fiori AM, Oliosi G, Pasquazzi L. *Compositional k-prototypes: a proposal for CoDa clustering with zeros*
10:20 – 10:40	Comas-Cufí M, Martín-Fernández JA, Lama Zubiran P de la. *Local search heuristics for balance optimisation in compositional data analysis*
10:40 – 11:00	Czolková A, Hron K, Greven S. *Functional principal component analysis of multivariate densities in Bayes spaces*
11:00 – 11:50	**Coffee-break**
Microbiology I	**chair: Mateu-Figueras G**
11:50 – 12:10	Silva-Solar S, Amann R, Knittel K *Ecological succession of bacterial communities on single sand grains*
12:10 – 12:30	Calle ML, Pujolassos M, Susin A *Bias of compositional microbiome differential abundance testing*
12:30 – 12:50	Te Beest DE, Mildau K, Engel B, Gort G, Lambert J, Swinkels SHN, Eeuwijk FA van *Pairwise ratio-based differential abundance analysis of infant microbiome 16S sequencing data*
12:50 – 14:30	**Lunch**
Keynote I:	**chair: Tolosana-Delgado R**
14:30 – 15:30	**Paul-Gauthier, N** ***Dealing with multiclass and multiple hypotheses problems with the Aitchison geometry of the simplex***

Earth Sciences I	chair: Tolosana-Delgado R
15:30 – 15:50	Gijtenbeek M van, McKinley J, Ruffell A *Using compositional data analysis to assess the applicability of geological databases as reference materials for urban forensic surface soil investigations*
15:50 – 16:10	Engle M, Venzor Nava A, Sanchez D *Assessing performance of compositional data analysis algorithms for the imputation of missing geochemical data to inform oil and gas wastewater re-use*
16:10 – 16:30	Valls Álvarez RA *Isometric log-ratio transformations for interpreting enzyme-leach geochemical signatures in volcanic terranes: insights from Hispaniola arc systems*
16:30 – 16:50	Wehner M *Using X-means and Imputation to Obtain Geologic Insights from Incomplete Inorganic Geochemical Data (XRF)*
16:50 – 17:10	Rezaei S, Molayemat H, Torab FM *A compositional approach to delineate fertile from barren veins in a gold mine*
17:10 – 17:30	
17:30 – 18:00	**Coffee-break**

Poster session	chair: Comas-Cufí M
18:00 – 19:00	
	Albuquerque T, Egozcue JJ, Pawlowsky-Glahn V, Carvalho P *Compositional singularity as a tool for mineral exploration strategy design* Almeida J *Compositional geostatistical mapping of carbonate lithological transitions using log-ratio coordinates* Araiz C, Erb I, Jin S, Grapotte M, Notredame C. *Inferring feature-level significance from pairwise ratios in differential abundance analysis* Kanjiradan Veetil A, Dilip TR. *Decoding the dynamics of female cancer mortality in India through compositional data analysis* Correia M, Pina P. *Evaluating machine learning performance in compositional landscapes: a case Study on Antarctic vegetation mapping* Browne TS, Edgell DR, Gloor GB. *Compositionality improves machine learning training for CRISPR cleavage prediction* Keivani F, Coenders G. *Adapting Altman's bankruptcy prediction model to the compositional data methodology* Fernández-Suárez J, Egozcue JJ, Pawlowsky-Glahn V. *Weighted bivariate density analysis of randomly sampled geochemical data* Monti GS, Mateu-Figueras G, Pawlowsky-Glahn V, Egozcue JJ *Detecting outliers in compositional data using the logratio t model* Ortego MI, Pantaleoni Reluy N, Lantada N *Who bears the loss? A compositional look at disaster payouts in Catalonia* Porro F, Rapallo F, Sommariva S *Test for the individuation of a common subspace in compositional datasets with structural zeros* Sloth Carlsen A, Chen T, Groves T, Luke Cowie N, Brinch Ch, Keld Nielsen L *Isotopologue distributions are compositions*

	Ríos Martín I, Creus Martí I, Santonja FJ *A Bayesian support vector machine approach for compositional data classification*
	Tang W, Fačevicová K, Nordhausen K, Taskinen S *A critical comparison of handling zeros in high dimensional compositional count data*
20:00 – 21:30	**Fado Night Dinner (only for 50)**

Wednesday

Health I	**chair: Calle ML**
09:00 – 09:20	K Kanjiradan Veetil A, Dilip TR *Integrating compositional data analysis in cancer epidemiology: evaluating the cancer transition in India*
09:20 – 09:40	McKinley JM, Räsänen-Young C, Mueller U, Mullineaux S, Hill C, McGuinness B, Hunter R, McKnight AJ *Exploring how our biology changes in response to environmental stimuli to detect biomarkers of renal disease: A compositional approach*
09:40 – 10:00	Miatke A, Smith A, Dumuid D, Stanford T et al *Socioeconomic status and cognitive function in later life: examining 24-h time-use composition as a potential mediator*
10:00 – 10:20	Morimoto J *Why nutrition data are compositional*
10:20 – 10:40	Narayana JK, Chotirmall SH *The cost of ignoring compositionality: CoDA unmasks treatment effects in COPD clinical trials*
10:40 – 11:00	Prinelli F *Animal and plant protein intake and all-cause mortality: A 24-year compositional analysis in a population-based cohort*
11:00 – 11:50	**Coffee-break**

Social Sciences	**chair: McKinley J**
11:50 – 12:10	Ferrer-Rosell B, Martin-Fuentes E, Petronella Swart M, Thirumaran K *When luxury derails: shifting patterns of tourist dissatisfaction on luxury trains*
12:10 – 12:30	Vega Baquero JD, Santolino M *Proportionality between allocations in asset management*
12:30 – 12:50	Yoshida T, Yamada I *Moran's I for compositional data: A measure of spatial auto- and cross-correlations*
12:50 – 14:30	**Lunch**
14:30 – 19:00	**Excursion: Old Town and University Library**

Thursday

Methods II	**chair: Ferrer-Rosell B**
09:00 – 09:20	Erb I, Ay N. *Quantifying scale-free information in pairwise associations of variables for counts and density data*

09:20 – 09:40	Eilers, P *Generalised TrioScale for transformation and modelling of compositional data*
09:40 – 10:00	Tolosana-Delgado R *Subcompositions-based tree methods*
10:00 – 10:20	Pavlů I, Tolosana-Delgado R, Templ M, Boogaart KG van den *Analysing distributional and density data using the Bayes space framework in R*
10:20 – 10:40	Saperas-Riera J, Lama Zubiran P de la, Martín-Fernández JA *Some thougths on L1-type penalisation norms in convex clustering for compositional data*
10:40 – 11:00	Sebatjane P, Lubbe, le Roux N *Calibrating centered log-ratio biplots*
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Health II	**chair: Erb I**
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12:10 – 12:30	Stanford T, Mellow M, Robins B, Smith AE, Dumuid D *Personalised prediction of cognitive outcomes and traffic lights to signal positive changes in time-use composition*
12:30 – 12:50	Tang N, Huang J *First application of CoDA to the high-dimensional (HD) biological data matrices of single-cell RNA-sequencing (CoDA-hd R package)*
12:50 – 14:30	**Lunch**
Keynote II:	**chair: J Palarea**
14:30 – 15:30	**Genius Serra P** ***CoDa in neuroimaging studies: exploring its application to identify early brain changes in Alzheimer's Disease***
Methods III	**chair: J Palarea**
15:30 – 15:50	Fačevicová K, Rieser C, Filzmoser P *Cellwise robust covariance estimation for compositions*
15:50 – 16:10	Faugeras, OP *Log-free divergence and covariance matrix for compositional data I: The affine/barycentric approach*
16:10 – 16:30	Nakamura K *Visualizing symmetric and skew-symmetric interactions in square compositional tables*
16:30 – 16:50	Oviedo Caro MA, Stanford T, Miatke A et al *A trajectory-based clustering approach to intra-week time-use compositions*
16:50 – 17:10	Škorňa S, Machalová J *Spline approximation of probability density functions*
17:10 – 17:30	Ricciotti L, Giordano S, Maruotti A *A latent class approach for dynamic clustering of compositional data*
17:30 – 18:20	**Coffee-break**
18:20 – 19:00	**General Assembly CoDa-Association**
19:00 – 21:30	**CoDaWork-dinner**

Friday

Methods IV	**chair: Gloor GB**	
09:00 – 09:20	Mondon C, Thi Trinh H, Martín-Fernández JA, Thomas-Agnan C *Testing mean densities with an application to climate change in Vietnam*	
09:20 – 09:40	Higgins P *Compositional monitoring of energy-mix drift on the simplex*	
Microbiology II	**chair: Gloor GB**	
09:40 – 10:00	Santos SJ, Murariu AC, Silverman J, Gloor GB *Accounting for scale controls for false discovery rate in the analysis of high throughput sequencing data*	
10:00 – 10:20	Araiz C, Erb I, Jin S, Grapotte M, Notredame C *Inferring feature-level significance from pairwise ratios in differential abundance analysis*	
10:20 – 10:40	Galván-Femenía I, Palarea-Albaladejo J. *Compositional modelling of mutational signatures for survival prediction in cancer*	
10:40 – 11:00	Creus-Martí I, Comas-Cufí M, Palarea-Albaladejo J *Modelling longitudinal microbiome data subject to time-varying treatment regime through low-dimensional projection*	
11:00 – 11:50	**Coffee-break**	
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CoDa in neuroimaging studies: exploring its application to identify early brain changes in Alzheimer's Disease

Patricia Genius Serra
Barcelonaβeta Brain Research Center in Barcelona. Spain.

In neuroimaging studies, common multivariate methods rely on pairwise covariances between regions, overlooking the broader compositional and bounded nature of the brain. In neurodegenerative diseases, such as Alzheimer's disease (AD), several neuroimaging measures have been widely described as intermediate phenotypes, which are measurable components in the pathway between the clinical phenotype and the genotype, and emerge as powerful tools for gene discovery. Recent research highlights the value of compositional approaches in brain imaging studies, particularly in AD, where structural alterations emerge years before symptoms onset and changes in one region must be interpreted in the context of changes occurring across interconnected regions. The current study aims to identify relative brain volumetric variations in cortical and subcortical regions that (i) are associated with different AD stages along the disease continuum and (ii) vary by AD genetic risk. Using compositional data analysis (CoDA) we identified the relative variation of brain regions' volumes, captured in compositional brain scores, associated with specific disease stages along the disease continuum. The genetic profile for AD shaped the way that these structures were interdependently interacting all along the continuum. CoDA provides an accurate analysis of brain imaging data accounting for the common overlooked compositional nature of the brain.

Keywords: neuroimaging measures, Alzheimer's disease, gene discovery.

Presenter: Patricia Genius Serra, pgenius@barcelonabeta.org

Dealing with multiclass and multiple hypotheses problems with the Aitchison geometry of the simplex

Noé Paul-Gauthier
CNRS Laboratoire d'Informatique et des Systèmes, Marseille, France.

Discrete probability distributions are key elements in statistics and machine learning. In particular, classification models and probabilistic prediction produce probability distributions over a finite set of possible classes or hypotheses. While the Aitchison geometry of the simplex has been actively promoted for the analysis of compositional data, and despite the neat result that perturbation is the Bayes' rule, the use of the Aitchison geometry on the probability simplex has been less discussed in the context of statistical inference and machine learning. In this talk, we will argue that the linear structure provided by the Aitchison geometry of the simplex allows us to tackle multiclass problems going beyond the standard one-vs-one and one-vs-rest approaches. In particular, we will lean on two practical examples. First, we will see how the Aitchison geometry can be used to extend results on the calibration of probability distributions and likelihood functions from the binary to the multiclass case. Secondly, we will see how the Aitchison geometry can be used to extend the concept of Shapley value to the simplex for explaining probabilistic predictions. With these two practical yet very distinct examples, we aim to demonstrate the wide applicability and relevance of the Aitchison geometry on the probability simplex in statistics and machine learning.

Keywords: Aitchison geometry, machine learning, calibration, Shapley value.

Presenter: Noé Paul-Gauthier, paul-gauthier.noe@lis-lab.fr

Compositional Singularity as a Tool for Mineral Exploration Strategy Design

Teresa Albuquerque[1], Juan José Egozcue[2],
Vera Pawlowsky-Glahn[3], Paula Carvalho[4]

[1]CERNAS—Instituto Politécnico de Castelo Branco, Polytechnic University, Portugal;
[2]Technical University of Catalonia, Barcelona, Spain;
[3]University of Girona, Girona, Spain;
[4]Universidade de Aveiro, Aveiro, Portugal.

The Valongo Anticline contains multiple Sb–Au and As–Au mineral deposits, many of which have been mined since Roman times but are now abandoned. To facilitate future exploration efforts, the Baixo-Douro consortium (EDM– ECD–BRGM) conducted a stream-sediment sampling campaign, collecting 801 samples at an average density of 4 per square kilometre. The geochemical data, collected on a systematic spatial grid, were initially analysed through univariate spatial singularity analysis to identify potential zones of mineralization. Given that geochemical data are compositional, a compositional singularity assessment is necessary. This methodology employs procedures analogous to traditional techniques but utilizes log-ratio transformations—such as isometric or centered log-ratios— instead of raw concentration values. Determining promising geological units and potential Au-Sb hotspots using the concept of compositional singularity is a valuable tool for enhancing mineral exploration in the study. The probability of identifying significant clusters in subsequent explorations is modelled using Simplicial Indicator Kriging. For demonstration purposes, singularities for Au and Sb were calculated and examined, with thresholds established based on samples exhibiting high-concentration, high-singularity patterns. The analysis is assessed using combinations of low concentration–high singularity and high concentration–low singularity as proxies.

Keywords: Mineral Exploration, Compositional Singularity; Simplicial Indicator Kriging

Presenter: Teresa Albuquerque, teresal@ipcb.pt

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Compositional geostatistical mapping of carbonate lithological transitions using log-ratio coordinates

José Almeida

GeoBioTec and Earth Sciences Department, NOVA-FCT, Campus Caparica, 2829-516 Caparica, Portugal.

The spatial discrimination of carbonate lithologies, such as limestone, marl, and dolomite, requires explicit consideration of the compositional nature of geochemical data. In this work, a compositional geostatistical framework is applied to map lithological transitions using five-part geochemical compositions (Si, Fe, Al, Ca, Mg), constrained to a constant sum and defined on the Simplex, where only relative information is relevant and absolute concentrations are not meaningful. The data were expressed in isometric log-ratio (ilr) coordinates using balances with direct geochemical interpretation. These balances represent the main controls on lithological variability: (i) the contrast between carbonate (Ca, Mg) and detrital (Si, Al, Fe) components, associated with the transition from marl to more carbonate-rich lithologies, and (ii) the contrast between Ca and Mg, reflecting the calcite–dolomite continuum. Together, these two balances capture the dominant geochemical gradients controlling lithological variability across the study area. Spatial structure was characterised using variogram analysis of the ilr coordinates, followed by cokriging interpolation, enabling joint spatial modelling of the correlated balances. The resulting maps of ilr coordinates are interpreted directly as spatial expressions of lithological contrasts, avoiding back-transformation to the original components and preserving the compositional integrity of the data. In this setting, the first balance highlights the carbonate–marl transition, while the second captures the calcite–dolomite gradient. This approach provides a consistent and robust framework for representing gradual lithological transitions in limestone quarries for cement manufacturing, demonstrating that spatial patterns are more clearly resolved by compositional contrasts than by individual component maps, thereby improving the geological interpretability of the results.

Keywords: Compositional data analysis; ilr coordinates; geostatistics; carbonate quarry, carbonate lithologies

Presenter: José Almeida, ja@fct.unl.pt

Inferring feature-level significance from pairwise ratios in differential abundance analysis

Cristina Araiz[1], Ionas Erb[1], Suzanne Jin[1],
Mathys Grapotte[1], Cédric Notredame[1]

[1]Centre for Genomic Regulation (CRG), Barcelona, Spain;

High-throughput sequencing experiments generate count data that quantify molecular features such as genes. These counts are constrained by sequencing depth and other global effects, making them intrinsically compositional. Differential analysis is routinely used to identify features associated with experimental conditions, yet standard workflows rely on normalization procedures whose validity depends on assumptions about the underlying biological context. Compositional data analysis offers a principled alternative by operating directly on pairwise ratios, thus avoiding an explicit reference while preserving the relative information encoded in the data. In genomics, differential abundance of pairwise logratios can replace the search for individually changing features. Changes in relative proportions across conditions reflect altered stoichiometries rather than absolute abundance. Despite their conceptual appeal, ratio-based methods have limited adoption because pairwise results are hard to interpret, computationally demanding at scale, and poorly aligned with downstream analyses that expect feature-wise statistics. We present a general strategy to infer feature-level significance from pairwise compositional signals while preserving the reference-free nature of the analysis. Inspired by Gene Set Enrichment Analysis (GSEA), we rank all gene pairs by a differential proportionality measure and, for each feature, treat the collection of pairs involving that feature as the set of interest. We then test whether those pairs are uniformly distributed along the ranked list or concentrated toward the extremes, yielding a feature-wise score and an associated significance measure. A methodological challenge is that, unlike classical enrichment settings, the ranked elements here are gene pairs, which are not independent. To address this, we introduce a subsampling scheme that computes enrichment statistics on carefully constructed subsets of pairs that preserve independence across features. Under the assumption of redundancy in pairwise information, this enables statistically valid estimation of feature-wise significance. It removes the need of user-defined thresholds, bypasses permutation-based false discovery rate estimation, and produces feature-wise metrics directly comparable to conventional differential analysis. Benchmarks across multiple sequencing datasets show that our approach recovers rankings similar to graph-based connectivity while improving interpretability and reducing computation time.

Keywords: differential abundance analysis, differential proportionality, normalisation-free analysis of sequencing data

Presenter: Cristina Araiz, cristina.araiz@crg.eu

Compositional k-prototypes: a proposal for CoDa clustering with zeros

Roberto Ascari[1], Anna Maria Fiori[1], Giulia Oliosi[1], Leo Pasquazzi[1]

[1]University of Milano-Bicocca, Italy.

Dissimilarity measures for clustering compositional data (CoDa) must be coherent with the relative nature of compositional information, as formalised through Aitchison's principles for log-ratio analysis. These measures require complete compositions (with strictly positive values in all parts), but, in practice, many data sets contain zero elements arising from multiple and different circumstances or processes. While count and rounded zeros are induced by limitations in experimental design or measuring instruments, essential zeros reflect the genuine absence of one or more components. Replacement strategies available in specialised literature for missing and censored data are therefore inappropriate for essential zeros, which are inherent to the data-generating process. Since zero patterns are typically excluded from the definition of compositional dissimilarities, clustering irregular CoDa samples (with zeros in one or more parts) poses considerable challenges. This work proposes a novel strategy for handling zeros of any type in partitional clustering of irregular CoDa under a not-missing-at-random (NMAR) pattern. We introduce a dissimilarity measure specifically designed for compositions where zeros are concentrated in one or few parts, redefining the latter through categorical variables with two or more levels. The dissimilarity between a (possibly irregular) composition and a cluster prototype is computed by combining the Aitchison distance for the sub-composition of regular (i.e., strictly positive) parts with simple matching coefficients for the categorical features that replace the irregular parts from the original CoDa set. The resulting measure is embedded into a cost function that can be iteratively minimised using Huang's k-prototype algorithm for mixed (categorical and numeric) attributes (Huang 1997, 1998; Szepannek, 2018). Cluster prototypes are represented by mean values for clr-transformed sub-compositions, and mode values for categorical features corresponding to irregular parts. We illustrate the method with an empirical application to country-level data on energy mix composition, where essential zeros naturally appear in the nuclear power share of countries that prohibit this source.

Keywords: irregular CoDa, partitional clustering, mixed data, energy mix

Presenter: Anna Maria Fiori, anna.fiori@unimib.it

Compositionality improves machine learning training for CRISPR cleavage prediction

Tyler S Browne[1], David R Edgell[1], Gregory B. Gloor[1]

[1]Department of Biochemistry, Western University, London, Canada;

The CRISPR-Cas9 single-guide RNA (sgRNA) complex can, in theory, cut at any arbitrary site, but in practice, only a minority of predicted sites can be cut efficiently. The reason for this disconnect is unknown, and the prediction of high-activity cleavage sites is difficult. We have shown that the use of compositionally-appropriate differences between information can be used as the training data for machine learning (ML), and that doing so results in a significant improvement in prediction accuracy. Here, we will show that curating the input data to remove uninformative sgRNA sequences results in a further improvement in accuracy. Finally, we will show how including compositional uncertainty in the curated training data results in a substantial improvement in prediction accuracy. Surprisingly, the inclusion of compositional uncertainty in the training data has a larger effect on prediction accuracy than does the ML architecture. Finally, these models predict with an accuracy greater than 80% even when predicting against data generated from a different assay, and for data generated from a different species. Thus, the inclusion of compositional inputs in the training data results in ML models that are generalizable and accurate.

Keywords: CRISPR prediction, sgRNA, machine learning, uncertainty, data curation

Presenter: Gregory Gloor, ggloor@uwo.ca

Bias of compositional microbiome differential abundance testing

M.Luz Calle[1,3], Meritxell Pujolassos[1], Antoni Susin[2]

[1] University of Vic – Central University of Catalonia, Spain;;
[2] Universitat Politècnica de Catalunya-Barcelona Tech, Spain;.
[3] Institut de Recerca i Innovació en Ciències de la Vida i de la Salut a la Catalunya Central (IRIS-CC), Spain.

One of the most relevant aspects in microbiome studies is the identification of microbial species that are differentially abundant across conditions. However, the compositional nature of microbiome data makes differential abundance testing challenging. The interdependence among the parts of the composition leads to spurious associations when the abundances of each component are analysed separately in relation to a response variable, resulting in a high number of false positives in differential abundance tests. Due to the growing awareness of the challenges of compositional data analysis, log-ratio transformations, such as the additive log-ratio (alr) or the centre log-ratio (clr) transformations, have become increasingly popular in microbiome studies. However, these transformations are also not free from bias. In this work, we present the theoretical expressions of the bias associated with the use of proportions (total sum scaling) and with log-ratio transformations (alr and clr) for univariate differential abundance tests. Based on these theoretical results, we describe the conditions required for each method to be unbiased. We compare the magnitude of the bias of the three approaches in different scenarios and identify those situations in which the bias is negligible. The factors that most strongly influence the bias are the dimension of the composition, the proportion of differentially abundant variables, and the distribution of relative abundances. Finally, assuming that most of the components have a null effect, we propose a heuristic method to obtain a suitable alr reference component that yields an unbiased estimate or one with negligible bias.

Keywords: log-ratio transformations, bias, differential abundance testing, microbiome compositions

Presenter: M.Luz Calle, malu.calle@uvic.cat

Local search heuristics for balance optimisation in compositional data analysis

Marc Comas-Cufí[1], Josep Antoni Martín-Fernández[1], Paula de la Lama Zubiran[1]

[1]Universitat de Girona.

In compositional data analysis, optimisation problems related to the construction of balances—such as principal balances or balance selection—are inherently combinatorial and defined over large, discrete, and highly structured search spaces. Consequently, the performance of heuristic local search methods in these settings depends critically on the definition of appropriate neighbourhood operations. In this work, we investigate specific neighbourhood operators for local search–based optimisation of balances, with a particular focus on the Tabu Search algorithm. Candidate solutions are represented as signed partitions of parts, and neighbourhood structures are designed to induce controlled modifications of their combinatorial configuration while preserving their validity as signed partitions. The proposed heuristics are evaluated in high-dimensional settings where exhaustive methods are infeasible. Specifically, we analyse performance across increasing dimensionality, focusing on the trade-off between solution quality and computational cost. Our results demonstrate that local search algorithms substantially reduce the computational cost of obtaining optimal, or near-optimal, solutions. Ultimately, this study provides practical guidelines for designing neighbourhood structures in heuristic optimisation methods for compositional data, with direct applications in statistics and machine learning.

Keywords: local-search, tabu-search, balance, neighbourhood

Presenter: Marc Comas-Cufí, marc.comas@udg.edu

Evaluating machine learning performance in compositional landscapes: a case Study on antarctic vegetation mapping

Miguel Correia[1], Pedro Pina[2]

[1]Earth Sciences Department, University of Coimbra, miguelcorreia2803@gmail.com;
[2]IDL-Coimbra and Earth Sciences Department, University of Coimbra, ppina@dct.uc.pt.

The classification of ice-free areas in the Antarctic Peninsula presents a unique challenge for statistical modelling due to the fragmented nature of vegetation and the inherent compositional properties of land cover data. In these environments, land cover types such as mosses, lichens, and bare geological surfaces (soils and rock outcrops) are often intermingled, leading to spectral mixing where the relative proportions of each component must sum to unity. This study explores the performance of three supervised machine learning algorithms – Support Vector Machines (SVM), Random Trees (RT), and k-Nearest Neighbours (kNN) – in classifying high and medium-resolution satellite imagery (WorldView-2, Sentinel-2, and Landsat 8) from the ice-free areas of Barton Peninsula in King George Island. We analysed the effectiveness of both pixel-based and object-based image analysis (OBIA) to handle the spatial dependence and compositional complexity of the landscape, namely, the relatively small dimension of the vegetation patches, to select the optimal threshold detection levels. Our findings reveal that kNN achieved the most robust results (Overall Accuracy of 0.91), particularly when dealing with the high-dimensional feature space of WorldView-2. This suggests that, in highly heterogeneous polar environments, local similarity-based classifiers may handle the non-linear boundaries between compositional classes more effectively than margin-based methods like SVM. The results are discussed in the context of classification uncertainty and the "mixed pixel" problem, providing insights into how machine learning can be optimized for monitoring sensitive ecosystems through a compositional lens.

Keywords: compositional data, remote sensing, machine learning, antarctic vegetation, kNN vs SVM

Presenter: Miguel Correia, miguelcorreia2803@gmail.com

Modelling longitudinal microbiome data subject to time-varying treatment regime through low-dimensional projection

Irene Creus-Martí[1], Marc Comas-Cufí[2], Javier Palarea-Albaladejo[2]

[1]Department of Statistics and Operational Research, Universitat de València;
[2]Department of Computer Science, Applied Mathematics and Statistics, University of Girona.

Current research suggests that the temporal dynamics of the gut microbiome is intimately tied to the health of the host species. Thus, disruptions to its temporal equilibrium are commonly linked to pathological states, while the reestablishment of a steady microbial arrangement often leads to clinical recovery. Moreover, recent studies have reported seasonal fluctuations in the gut microbiota, emphasising the relevance of considering the temporal dimension to anticipate shifts in its microbial composition. In this context, we present an analytical approach that helps to capture the underlying patterns of temporal variation and ease their interpretation from complex longitudinal microbiome observations, simultaneously accommodating compositional aspects and the inherent features of temporal data. In brief, a low-dimensional representation of the data through principal component analysis allows for summarising the dominant variation patterns and identifying their key drivers within the composition. Then, mixed-effects modelling allows to investigate the potential effects of time-varying covariates. The method is illustrated using a longitudinal dataset including microbiota samples of German cockroach *(Blattella germanica)* exposed to intermittent administration of Kanamycin antibiotic treatment over time, along with those from an untreated control group. The results demonstrate the ability of the proposal to synthesise the information and differentiate both treatment regimes over time and treatment groups, while facilitating the understanding of the dynamics of the microbiome composition and the identification of the leading taxa involved.

Keywords: modelling, longitudinal data, time-varying treatment, microbiome

Presenter: Irene Creus-Martí, irene.creus@uv.es

Functional principal component analysis of multivariate densities in Bayes spaces

Adéla Czolková[1], Karel Hron[1], Sonja Greven[2]

[1]Palacky University Olomouc, Czech Republic;
[2]Humboldt-Universität zu Berlin, Germany.

The Bayes space provides a Hilbert space structure for analysing probability density functions (PDFs), equipping them with a geometry that reflects their relative and constrained nature. A key tool in this framework is the centred logratio (clr) transformation, which establishes an isometric isomorphism between the Bayes space and the classical L^2 space. This makes it possible to apply functional data analysis (FDA) techniques, particularly functional principal component analysis (FPCA), to density data in the context of dimension reduction. For multivariate PDFs, embedding them in the Bayes space enables an orthogonal decomposition into independent and interactive components. Furthermore, the independent part can be decomposed into mutually orthogonal geometric marginals. This structure provides more profound insights into the sources of variation in multivariate densities. It will be shown that applying FPCA directly to multivariate densities is equivalent in a certain sense to applying multivariate FPCA to their decomposed form, with the resulting eigenfunctions and scores decomposing accordingly. The unique decomposition based on these theoretical results was applied to empirical housing data, demonstrating the interpretability and practical value of this approach. The results of this application will be presented as well.

Keywords: probability density functions, orthogonal decomposition, functional principal component analysis

Presenter: Adéla Czolková, adela.czolkova01@upol.cz

Accounting for scale controls for false discovery rate in the analysis of high throughput sequencing data

Scott J. Dos Santos[1], Andreea C. Murariu[1], Justin D. Silverman[2], Gregory B. Gloor[1]

[1]Department of Biochemistry, The University of Western Ontario, London, Canada;
[2]Department of Statistics, Pennsylvania State University, PA, United States.

All approaches for the analysis of high throughput sequencing analysis make implicit, but hidden, assumptions about the scale (absolute abundance) of the environment being sampled. This misspecification can result in both false positive and false negative inference when conducting differential abundance/differential expression (DA/DE) analysis. False discovery rate (FDR) control is a known problem in DA/DE analysis, and the current approach of using a dual cutoff (low p-value and large difference) is known not be an appropriate solution. Different tools employ distinct normalisations to attempt to correct for underlying variation when analyzing the count data from high throughput sequencing data; that is, normalisations, including CoDa-based ones, make incorrect assumptions about the underlying scale (i.e. size) of the biological system. While we usually do not know the underlying true scale, we can as a minimum include uncertainty of scale estimation. We used 10 different RNA-seq datasets and showed that scale misspecification, by both compositional and non-compositional DA/DE tools, caused poor FDR control. We show that the FDR can be controlled appropriately when uncertainty is included in the scale model by modifying the CoDa normalizations used by ALDEx2 or ALDEx3. This can be implemented simply and with essentially no computational overhead. We show that the inclusion of scale uncertainty in the analysis results in a dynamic relationship between the observed p-value and the difference between groups needed to obtain that p-value. Furthermore, we show a predictable relationship between the amount of scale uncertainty and the incremental increase in the minimum difference between groups that is required to be classed as significant. Overall, this work extends our understanding of the influence of absolute abundance when analyzing high throughput sequencing data despite its compositional and partially-specified nature. Finally, this work highlights the choice between sensitivity and false-discovery rate control that all researchers must make when analysing sequencing data.

Keywords: high throughput sequencing, absolute abundance, scale, Bayesian, false discovery rate

Presenter: Gregory B. Gloor, ggloor@uwo.ca

Generalized TrioScale for transformation and modelling of compositional data

Paul Eilers[1]

[1]Erasmus University Medical Center; Rotterdam, The Netherlands

At previous CoDaWork conferences, Mark de Rooij and I proposed TrioScale as an alternative to the ternary diagram for displaying compositional triples. The idea is to have three oblique axes, at angles of 0, 120 and 240 degrees through the origin, corresponding to the three possible log-ratios. To each data triple corresponds a point in the plane such that its projections on the three axes give the proper log-ratios. Transforming a collection of triples in this way generates a cloud of points. They are not restricted in any way, unlike the simplex for the ternary diagram. The construction is generalised to higher dimensions, mapping logratios of compositions with n components to an unrestricted Euclidean space with $n-1$ dimensions. This opens many opportunities for advanced statistical analysis. Examples are regression on compositions, principal components analysis, time series analysis and spatial smoothing. We are not limited to compositional triples, so I propose CompoScale as a better name. I will present theory and several applications, using an R package for computations and graphical display.

Keywords: transformation, Euclidean space, visualization

Presenter: Paul Eilers, p.eilers@erasmusmc.nl

Assessing performance of compositional data analysis algorithms for the imputation of missing geochemical data to inform oil and gas wastewater re-use

Mark Engle[1], Aldo Venzor Nava[1], and David Sanchez[1]

[1] University of Texas at El Paso, USA.

Produced water–deep groundwater and injected fluids–, co-generated from hydrocarbon reservoirs, is the most voluminous by-product of oil and gas extraction. Because local subsurface disposal options are limited, treating and reusing produced water is a promising alternative. However, designing effective treatment systems and use strategies requires comprehensive chemical data. Publicly available produced water datasets often have many missing or censored values, which makes it difficult to create well-informed management decisions. The objective of this project was to assess the performance of standard R-based compositional data analysis methods (from the robCompositions and zCompositions packages) to impute these missing concentration data, improving dataset completeness and enabling more reliable treatment decision-making.

Two pre-existing datasets were subset for Appalachian Basin shale reservoir data and combined with data for 20 commingled produced water samples. Initial data cleaning removed samples and features with $> 70\%$ missing values, prioritising completeness, leading to a dataset of 1 075 data points. Four traditional compositions data analysis methods were tested: K-Nearest Neighbours (KNN) of orthonormal log-ratio (olr) transformed data, ordinary least-squares regression-based imputation on olr transformed data (impCoda-lm), least-trimmed squares regression-based imputation on olr transformed data (impCoda-lts), and log-ratio Expectation-Maximisation (lrEM) methods. To assess the performance of the models, 10% of the data in the training dataset was masked. In general, imputation models predicted values within the correct order of magnitude but struggled with the prediction of trace elements, which are fewer in number and exhibit larger log variance than the major elements. The best performance, as assessed by the root mean square error (RMSE results), was found using the KNN method for major elements and the impCoda-lm method for some of the trace elements (Cd, Cr, Co, Cu, H+, and Ni). The robust least-trimmed squares regression method (impCoda-lts) had by far the worst prediction performance for major elements, but performed fairly well in the prediction of most of the trace elements. Examination of the mean absolute percent error (MAPE) results, which, unlike RMSE, is magnitude independent, showed that errors were roughly inversely proportional to concentration magnitude. KNN again showed the best performance using this metric, showing that predictions were less than an order of magnitude off for all elements except for Co and Mn. Such findings are typical for or better than trace element prediction using other machine learning models. These findings underscore the importance of balancing algorithmic complexity with dataset quality and domain-specific constraints. Critically, even imperfect imputations can inform treatment feasibility assessments, provided error margins ($\pm 20\%$) are acknowledged.

Keywords: imputation, produced waters, compositional Data

Presenter: Mark A. Engle, maengle@utep.edu

Quantifying scale-free information in pairwise associations of variables for counts and density data

Ionas Erb[1], Nihat Ay[2]

[1]Centre for Genomic Regulation (CRG), Barcelona, Spain;
[2]Institute for Data Science Foundations, Hamburg University of Technology, Germany.

One way of avoiding compositional bias in data is to discard those parts of them that correspond to size and only work with their shape. When studying variable association, we therefore want to use coefficients that are scale-free in some sense. While for data matrices a truly scale-free (but imperfect) measure of association is logratio variance, for cross-classified counts in the form of contingency tables, scale-free associations can be quantified using (similarly imperfect) cross-product ratios. Here we set out to define more useful scale-free quantities for pairwise association using a geometric reasoning. The CoDA approach to both contingency tables and bivariate density data has made use of so-called geometric marginals, which are obtained via marginalisation of CLR-transformed joint probabilities. It was shown that with their help, joint probabilities can be decomposed into a factorized part and an interaction part, corresponding to a part in the Euclidean CLR space that is projected onto the plane of factorizable distributions and a remaining orthogonal part. Here we propose an alternative, information-geometric approach based on the non-Euclidean Fisher-Rao metric. Using an information projection of a distribution onto the plane of factorizable distributions, the projection is the product of marginals (not of the geometric margins like in the Euclidean case), so the correct independent part in a probabilistic sense. We show that the remaining interaction part coincides with the one obtained from Euclidean projection. However, pure interaction without main effects only means equidistribution of geometric margins, and not necessarily that the marginals are equidistributed (which we need to have scale-freeness). Via another information projection, the formalism allows to define a marginal-free part of the distribution that, under certain conditions, coincides with the interaction part. For contingency tables, the marginal-free part can be obtained via iterative proportional fitting (IPF, a.k.a. the Sinkhorn algorithm), and in the binary case (2x2 tables), it always coincides with the interaction part. We propose to use this marginal-free part of the distribution to define a scale-free mutual information. To handle zeros, we also propose to make use of the log-free alpha-divergences within the information-geometric formalism. On a more applied level, recasting data matrices of counts as pairwise presence/absence contingency tables, our approach allows to define truly scale-free correlation coefficients between pairs of variables as well as their scale-free mutual information, with potential genomics applications such as gene-gene association in single-cell sequencing data.

Keywords: pairwise association, information geometry, sequencing data

Presenter: Ionas Erb, ionas.erb@crg.eu

Cellwise robust covariance estimation for compositions

Kamila Fačevicová[1], Christopher Rieser[2], Peter Filzmoser[2]

[1]Palacký University Olomouc, Czech Republic
[2]TU Wien, Austria.

Cellwise outliers are outliers in single entries of a compositional data matrix, and they can lead to a certain bias in the statistical analysis. Traditional casewise robust methods downweight entire outlying observations for the estimation, regardless of how many or which cells of an observation are contaminated. Cellwise robustness still makes use of the information contained in non-contaminated cells. The contribution introduces a novel cellwise robust estimator of the covariance structure of compositional data. The estimator is based on the Qn estimator applied to the entries of the variation matrix, which can be subsequently transformed into the clr covariance matrix. To preserve the theoretical properties of a covariance matrix, the inital estimate is replaced by its nearest positive semidefinite matrix. The performance of the proposed estimators is evaluated against classical and casewise robust alternatives through simulation studies, which show a promising results in terms of reconstructing the covariance structure under cellwise contamination. Additionaly, the application of the estimator to geochemical data provides a practical comparison of classical, rowwise robust and cellwise robust estimators.

Keywords: cellwise outliers, variation matrix, clr covariance matrix

Presenter: Kamila Fačevicová, kamila.facevicova@upol.cz

Log-free divergence and covariance matrix for compositional data I: The affine/barycentric approach

Olivier P. Faugeras[1]

[1]Toulouse School of Economics, Université de Toulouse Capitole, France.

Motivated by a way to overcome the issue of zeros in Aitchison's classical log-ratio analysis, we study the full CoDa simplex from the perspective of affine geometry. This view amounts to regarding CoDa as points (and not vectors), naturally expressed in barycentric coordinates. A decomposition formula for the displacement vector of two CoDa points yields a novel family of barycentric dissimilarity measures. In turn, these barycentric divergences allow to define (i) Fréchet means and their variants, (ii) isotropic and anisotropic analogues of the Gaussian distribution, and importantly (iii) variance and covariance matrices. All together, the new tools introduced in this talk provide a log-free, direct and unified way to deal with the whole CoDa space, exploiting the linear affine structure of CoDa, and effectively handling zeroes. A strikingly related approach based on the projective viewpoint and the exterior product is studied in a separate companion paper of the author.

Keywords: affine geometry, barycentric coordinates, barycentric divergence, barycentric variation matrix

Presenter: Olivier P. Faugeras, olivier.faugeras@tse-fr.eu

Weighted bivariate density analysis of randomly sampled geochemical data

Javier Fernández-Suárez[1], Juan José Egozcue[2], Vera Pawlowsky-Glahn[3]

[1]Complutense University and IGEO-CSIC, Madrid, Spain;
[2]Technical University of Catalonia, Barcelona, Spain;
[3]University of Girona, Girona, Spain.

Discriminating the provenance of detrital zircon populations in sedimentary rocks and depositional environments often relies solely on U–Pb detrital zircon ages. Although these ages provide essential chronological constraints on provenance, they commonly yield non-unique solutions. Integrating key secondary geochemical dimensions, such as initial Hafnium isotopic ratios, is therefore highly desirable, as these parameters can reveal dissimilarities that age data alone may fail to detect. This integrated U–Pb–Hf approach is expected to strengthen our ability to trace ancient and modern sediment pathways. It identifies isotopically distinct source regions and informs on the relative contributions of juvenile and evolved crustal material, thereby providing a richer geological framework for provenance interpretation. To perform the bivariate analysis, we applied two distinct compositional approaches, both of which yield weighted (with respect to an adaptive reference measure) inter-sample Aitchison distances within the framework of Bayes-Hilbert Spaces (BHS). These approaches are: (1) bivariate kernel density estimation, so that a density in BHS represents each sample; (2) dimensionality reduction through clustering of all sample data, yielding a composition for each sample. In both cases, bootstrapping was employed to account for sampling uncertainty in the multidimensional scaling (MDS) plots and compositional biplots. These procedures were applied to data from an illustrative geological case study. The methodologies preserve the coupled bivariate information and ensure that the calculated inter-sample distances — visualised through MDS plots and/or compositional biplots — remain geologically meaningful and interpretable in terms of stratigraphic, tectonic, and paleogeographic constraints. The proposed integrated geochronological–isotopic approach can identify finer-scale differences among zircon populations, thereby demonstrating the added value of the Hf isotopic ratio dimension for provenance analysis and providing a methodology that can be further generalised to multidimensional geochemical studies. Concerning geological interpretation, no core contradictions are observed between the kernel density and clustering approaches. However, the clustering approach shows the following advantages: i) a better visual intuition of intersample Aitchison distances, ii) the added advantage of the compositional biplot informing on the separate effect of U-Pb age and Hf isotopic ratio shifts, and iii) this approach largely overcomes the lack of robustness of kernel density estimation.

Keywords: Bayes-Hilbert space, reference measure, Aitchison distance, multidimensional scaling, detrital zircon

Presenter: Javier Fernández-Suárez, jfsuarez@geo.ucm.es

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When luxury derails: shifting patterns of tourist dissatisfaction on luxury trains

Berta Ferrer-Rosell[1], Eva Martin-Fuentes[1], Magdalena Petronella Swart[2], K. Thirumaran[3]

[1]Universitat de Lleida, Spain;
[2] University of South Africa, South Africa;
[3] James Cook University, Singapore.

Although research on tourist satisfaction is exhaustive, it remains a popular area of investigation due to changes in tourist behaviours and attitudes. Luxury tourism, yet research on luxury trains in general, is limited, particularly from an experience perspective. It is therefore necessary to assess the factors that influence service satisfaction amongst luxury train travellers. A popular source of tourist information is user-generated content, where travellers seek unbiased and unregulated information. While online reviews have been widely used in the tourism industry, they have traditionally focused on hotels or attractions. Literature on trains has rarely discussed the emerging identification of inconspicuous consumption in luxury. The aim is to fill this notable gap. In terms of data and method, the luxury trains recognised as the world's top 25 trains (https://www.irtsociety.com/trains) were selected. Online reviews posted on TripAdvisor between 2002 and 2025 about the selected trains were gathered for analysis. Octoparse, a web scraping software, was employed to systematically download more than 18,000 reviews in English from the selected luxury trains. We treated the review scores (from 1 to 5) as an indicator of consumer satisfaction, where lower scores (1 and 2 bubbles) were indicative of dissatisfaction. Complaint topics were extracted from the lower-score reviews and categorised based on the topics proposed by previous studies on the luxury sector (e.g., infrastructures, staff, value for money). We considered the luxury trains as the unit of analysis, for which we have a composition formed by six parts (D=6) corresponding to each complaint topic. Once clr's were computed, given that the units of analysis (trains) were less than 30, and therefore it was not feasible to model, we carried out the analyses to highlight the compositional nature of the data: a CoDa biplot and cluster analysis were performed. Results show that the dominant complaint topics refer to the service and staff, the train's infrastructure, and the value for money. The least common complaint topics are those related to crowd management and safety issues. This study contributes to the growing body of applied CoDA research by demonstrating the value of compositional methods for interpreting complex behavioural data derived from online reviews.

Keywords: satisfaction, service, experience, luxury travel, consumer behaviour

Presenter: Berta Ferrer-Rosell, berta.ferrer@udl.cat

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Compositional modelling of mutational signatures for survival prediction in cancer

Iván Galván-Femenía[1], Javier Palarea-Albaladejo[1],

[1]Department of Computer Science, Applied Mathematics and Statistics, University of Girona, Girona, Spain.

In the study of the cancer genome, mutational signatures summarise the imprints of diverse genetic mutational processes accumulated over time. They are associated with environmental exposures such as smoking or ultraviolet radiation, as well as to endogenous deficiencies in DNA repair. The mutation profile for an individual tumour represents a characteristic combination of relative contributions of mutation types, which confers the data a compositional structure. The study of the prognostic value of mutational signatures in cancer survival modelling remains limited, and the compositional nature of the data has been ignored in recent analyses based on multivariate Cox regression modelling, where signatures are dichotomised at the median. Using data from 1,069 colorectal cancer patients, we implemented a logratio approach to analyse more than 50 mutational signatures and evaluated their prognostic utility using elastic-net penalised Cox regression modelling. Moreover, incorporating a compositional risk score improved survival prediction, showing increased concordance index and hazard ratio in comparison to the ordinary approach applied to the raw mutational signatures. The results indicate that accounting for the relative structure of mutational signatures reveals associations with cancer survival that are undetected using standard modelling strategies. They have also motivated further methodological work on the estimation of mutational signatures using compositional divergence measures within the non-negative matrix factorisation algorithm, which is widely used for extracting mutational signatures in cancer studies. Overall, this work stresses the relevance and potential of the compositional methodology for the evaluation of mutational processes as prognostic biomarkers in oncology.

Keywords: mutational signatures, non-negative matrix factorisation, compositional divergence, survival analysis

Presenter: Iván Galván-Femenía, ivan.galvan@udg.edu

Using compositional data analysis to assess the applicability of geological databases as reference materials for urban forensic surface soil investigations

Mathijs van Gijtenbeek[1], Jennifer McKinley[1], Alastair Ruffell[1]

[1]Geography, School of the Natural Built Environment, Queen's University Belfast, Northern Ireland, UK.

Geological databases have demonstrated substantial value in forensics for both evidence and intelligence applications. However, the applicability of these geological databases for forensic purposes is constrained by inherent differences between geological and forensic methodologies and requirements, such as the sampling depth and spatial resolution. Geological databases are typically based on subsurface soil samples, whereas most forensic soil samples are collected from the surface, which are more susceptible to alteration through anthropogenic activities. These discrepancies are particularly evident in urban areas, which are characterised by increased anthropogenic activity. As a result, soils in urban areas are more prone to change, leading to a potential increase in spatial variability and a possible deviation from their original composition. This raises an important question: How does such variability impact the applicability of existing geological databases for forensic purposes? To address this, a multi-proxy approach was applied, including soil mineralogy obtained through automated SEMEDS analysis (QEMSCAN) and grain size distribution determined using laser diffraction particle size analysis. Both are commonly employed analytical techniques in geoforensics and produce compositional data. Despite this, compositional data analysis (CoDA) is not yet standard practice in the field. In this study, CoDA is central to interpreting spatial variability within each site and assessing the influence of anthropogenic activity. CoDA methods such as centred log-ratio transformations and biplots, and orthonormal log-ratio balance dendrograms were used to reveal intra-site spatial patterns and to distinguish forensic samples from those in the geological reference database.

Keywords: Geoforensics, clr-biplot, balance dendrogram, spatial variation

Presenter: Mathijs van Gijtenbeek, mvangijtenbeek01@qub.ac.uk

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Compositional monitoring of energy-mix drift on the simplex

Peter Higgins

Independent researcher, Markham, Ontario, Canada.

Monitoring frameworks in the environmental and governance literature typically compare observed values against external thresholds or conceptual models. None, to the author's knowledge, operates natively on compositions — tracking how the proportional allocation of a finite system evolves over time using the geometric structure of the simplex. This contribution proposes a monitoring protocol that treats energy-generation portfolios as compositional data and detects structural drift using tools from compositional data analysis. Each reporting period produces a composition in the simplex, representing the proportional share of electricity generated by each carrier (coal, gas, nuclear, hydro, solar, wind, and others). Change between periods is measured as perturbation — the compositional ratio between consecutive observations — rather than arithmetic difference, consistent with the Aitchison geometry of the sample space. The magnitude of period-to-period change is quantified using the Aitchison distance. The protocol is applied to publicly available electricity generation data from EMBER (CC BY 4.0) for Germany, Japan, and the United Kingdom over the period 2000–2025, with nine carrier types forming compositions on the 8-simplex. The resulting compositional time series reveal structural transitions that are invisible to non-compositional methods: Japan's post-Fukushima nuclear withdrawal appears as a large perturbation spike in 2011–2012; Germany's nuclear phase-out traces a continuous trajectory on the simplex toward the renewable vertex; the United Kingdom's coal exit registers as an abrupt regime change in the Aitchison distance matrix. A concentration measure related to effective diversity is computed alongside the compositional trajectory. The relationship between this measure and the Aitchison norm of the composition is noted as an open question for the community. The central claim is that no existing monitoring framework performs compositional change detection at the carrier level with formal perturbation-based drift measurement. This claim is presented as falsifiable: the contribution identifies four specific ways it could be defeated, and the author would welcome that outcome. All data, code, and documentation are publicly available.

Keywords: compositional time series, perturbation, drift detection, energy mix

Presenter: Peter Higgins, PeterHiggins@roguewaveaudio.com

Decoding the dynamics of female cancer mortality in India through compositional data analysis

Ananya Kanjiradan Veetil[1], T R Dilip[1]

[1]International Institute for Population Sciences, India.

Background: Cancer transition theory describes a shift from infection-related cancers to cancers associated with lifestyle and ageing as populations develop. Evidence for this transition among the Indian population remains limited, partly because most studies examine cancer sites separately and overlook the compositional nature of mortality data. This study applies a compositional approach to examine long-term changes in female cancer mortality across Indian regions.
Data and Methods: Annual female cancer mortality data for breast, cervix, and oral cancers from 1980 to 2021 were obtained from the Global Burden of Disease study. Indian states were grouped into six regions (Northeast, North, West, Central, South, and East), and analyses were conducted at the regional level. Mortality data were analysed as compositions using Compositional Data Analysis (CoDA), allowing a valid assessment of relative change. Temporal dynamics were explored using log-ratio transformations, and vector autoregressive models were used to characterise trends and generate region-specific projections.
Results: Across all regions, the relative contribution of breast cancer mortality increased steadily over time, and breast cancer overtook cervix as the leading cause of female cancer mortality at different time points across regions. Cervical cancer mortality declined consistently in all regions, although the pace of decline varied substantially. Oral cancer mortality showed marked regional heterogeneity, declining in some regions but remaining stable or increasing in others, particularly in the Northeast and Central regions. Projections to 2031 suggest that breast cancer will continue to dominate female cancer mortality in most regions, though the speed of transition differs across India.
Conclusion: Female cancer mortality in India is undergoing a compositional re-organisation rather than parallel site-specific changes. By focusing on relative cancer burdens, CoDA provides a clearer and more meaningful framework for understanding cancer transition, highlighting regional inequalities and offering policy-relevant insights for targeted cancer prevention, screening, and surveillance.

Keywords: VAR model, compositional time series analysis, projection

Presenter: A. Kanjiradan Veetil, kvananya1999@gmail.com

Integrating compositional data analysis in cancer epidemiology: evaluating the cancer transition in India

Ananya Kanjiradan Veetil[1], T R Dilip[1]

[1]International Institute for Population Sciences, India.

Background: India is undergoing rapid demographic and epidemiologic transitions, accompanied by a growing cancer burden. While cancer transitions, a shift from infection-related to lifestyle-related cancers, is observed in high-income countries, evidence for a similar transition in India remains limited. Existing studies focus on site-specific trends. Describing changes in individual cancers fails to capture how cancer types reconfigure relative to one another within the overall cancer burden. From a public-health policy perspective, such relative shifts are critical for prioritising prevention and resource-allocation. This study examines whether cancer patterns in urban India reflect a broader cancer transition when analysed as a compositional system.

Methods: Using population-based cancer registry data from Chennai (1982–2012), we analysed incidence patterns for nine cancers: breast, cervix, oral cavity, lung, colorectum, prostate, stomach, oesophagus, and liver. Initially, Joinpoint regression was used to assess site-specific temporal trends. Cancer transition was then examined using CoDA. Data were transformed into ilr-coordinates using Sequential Binary Partition to analyse relative shifts between cancer groups. Compositional linear regression with time as predictor and piecewise trend-filtering were employed to identify temporal trends and structural breaks in cancer dominance trajectories.

Results: Joinpoint analyses showed increasing trends for lifestyle-and tobacco-related cancers and declining trends for infection-related cancers, particularly cervix and stomach. CoDA regression demonstrated that time explained substantial compositional variability ($R^2 \approx 53\%$ males; $\approx 63\%$ females), indicating systematic reorganisation of cancer composition. Among males, oral cancer remained the leading site, with lung and colorectal gaining dominance and stomach and oesophagus declining, while among females, breast replaced cervix as leading cancer after the 2000s, alongside rising lung and colorectal cancers. Piecewise trend-filtering revealed distinct non-linear transition phases, particularly during the late 1980s and 1990s.

Conclusions: Cancer transition in India is best understood as a reorganisation of cancer composition. CoDA provides a useful analytic framework for tracking dominance shifts and informing cancer control priorities.

Keywords: CODA LRM, compositional trend filtering, cancer epidemiology

Presenter: A. Kanjiradan Veetil, kvananya1999@gmail.com

Adapting Altman's bankruptcy prediction model to the compositional data methodology

Fatemeh Keivani[1], Germa Coenders[1]

[1]University of Girona, Spain.

Using standard financial ratios as variables in statistical analyses has been related to several serious problems, such as extreme outliers, asymmetry, non-normality, non-linearity, and even dependence of the results on the arbitrary decision regarding which accounting figure appears in the numerator and which in the denominator of the ratio. The compositional-data methodology has been successfully applied to solve these problems and has always yielded substantially different results when compared to standard financial ratios. An under-researched area is that of the use of financial log-ratios computed with the compositional-data methodology to predict outcome variables such as bankruptcy or the related terms of business default, insolvency or failure. Another under-researched area is the use of machine learning methods in combination with compositional log-ratios. The present article adapts the classical Altman bankruptcy prediction model to the compositional methodology with pairwise log-ratios and three common statistical and machine learning tools: logistic regression models, k-nearest neighbours, and random forests. Data from the sector in the Spanish economy with the largest number of bankrupt firms according to the first two digits of the NACE code (46XX "wholesale trade, except of motor vehicles and motorcycles") were obtained from the Iberian Balance sheet Analysis System. The sample size (31,131 firms, of which 97 were bankrupt) was divided into a training and a validation dataset. The training data set was downsampled to one healthy firm to each bankrupt firm. No outliers were removed. Focusing on predictive performance, the results show that compositional methods are better in terms of sensitivity, with mixed results regarding specificity, compositional random forests and compositional logistic regression behaving the best.

Keywords: Machine learning, bankruptcy, outliers, accounting ratios

Presenter: Germa Coenders, germa.coenders@udg.edu

The cost of ignoring compositionality: CoDA unmasks treatment effects in COPD clinical trials

Jayanth Kumar Narayana[1], Sanjay H. Chotirmall[2],

[1]Lee Kong Chian School of Medicine, Nanyang Technological University, Singapore;
[2]Department of Respiratory and Critical Care Medicine, Tan Tock Seng Hospital, Singapore.

Introduction: COPD affects over 400 million people globally, with bacterial infections causing 50% of exacerbations. Despite the lung microbiome's critical role in disease progression, microbiome dynamics during therapeutic interventions remain poorly characterised in clinical trials. Microbiome data are inherently compositional, yet translational studies typically ignore this constraint, relying on distance-based metrics (e.g. Bray-Curtis dissimilarity) that compress high-dimensional data into univariate measures with reduced statistical power. While CoDA methods exist, they remain underutilized in clinical research. We demonstrate how compositional approaches reveal antibiotic treatment effects completely masked by standard methods.

Methods: We analysed sputum metagenomes ($n = 93$ patients; 256 samples over 52 weeks) from a phase III randomized, double-blind, placebo-controlled trial (NCT02305940) comparing daily doxycycline (100mg) versus placebo, with sampling at baseline and 3-month intervals. After prevalence filtering ($\geq 20\%$) and multiplicative zero replacement, we compared standard methods (diversity metrics, Bray-Curtis dissimilarity, PERMANOVA) with compositional approaches: inverse perturbation ($\ominus$) to calculate compositional change residuals, Aitchison norm for quantifying change magnitude, and ALDEx2 for differential abundance testing. Longitudinal dynamics were assessed through moving-window analysis between consecutive visits.

Results: Standard approaches (diversity indices, ecological evenness, Bray-Curtis dissimilarity, PERMANOVA) detected no significant differences between doxycycline and placebo groups at baseline or trial-end ($p > 0.05$), suggesting no treatment effect. In stark contrast, compositional analysis using inverse perturbation (trial-end $\ominus$ baseline) revealed significant treatment-induced changes. The active arm exhibited notable shifts in Streptococcus and Veillonella, with overall compositional change magnitude (Aitchison norm) significantly exceeding placebo ($p < 0.05$)-treatment effects completely masked by distance-based metrics. Moving-window analysis revealed time-dependent effects with three distinct temporal patterns: consistent changes (Rothia throughout treatment), short-term effects (Burkholderia, $0-3$ months), and long-term effects (Moraxella, $9-12$ months).

Conclusion: Classical subtraction operations are mathematically inappropriate for compositional data and fail to account for baseline heterogeneity. Perturbation-based methods properly quantify compositional change while preserving interpretability, revealing treatment effects and temporal dynamics masked by distance-based approaches. This demonstrates CoDA's essential role in longitudinal microbiome studies and clinical trial design.

Keywords: microbiome, respiratory

Presenter: Jayanth Kumar Narayana, jayanth.narayana@ntu.edu.sg

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Exploring how our biology changes in response to environmental stimuli to detect biomarkers of renal disease: A compositional approach

Jennifer M McKinley[1], Carys Räsänen-Young[2], U. Mueller[3] Shay Mullineaux[2], Claire Hill[2], Bernadette McGuinness[2], Ruth Hunter[2], Amy Jayne McKnight[2]

[1]Geography, School of Natural and Built Environment, Queen's University Belfast, Belfast, United Kingdom;
[2]Centre for Public Health, School of Medicine, Dentistry and Biomedical Sciences, Queen's University Belfast, Belfast, United Kingdom;
[3]School of Science, Edith Cowan University, Perth, Western Australia.

Exposure analysis undertaken as part of the UKRI ESRC funded SPACE ('Supporting Environments for Physical and Social Activity, Healthy Ageing and Cognitive Health') using the Northern Ireland Cohort for the Longitudinal Study of Ageing (NICOLA) cohort, comprising 8,000 older people, allows us to explore how different environmental factors relate to our health. Chronic Kidney Disease (CKD), a collective term for many causes of progressive renal failure, is also increasing worldwide due to ageing, obesity & diabetes. In this study, we use analysed element concentrations in urine samples, adjusted for urine creatinine, from a subset (247 persons) of the NICOLA cohort to explore how our biology changes in response to environmental stimuli to detect biomarkers of renal disease. A compositional balance approach is used to assess the association of 18 element concentrations (As, B, Ca, Cd, Co, Cu, Fe, Li, Mg, Mo, P, Rb, S, Se, Si, Sr, Ti and Zn) in urine with estimated Glomerular Filtration Rate (eGFR) and the Albumin-Creatinine Ratio (ACR). Results of the forward-selection method using the selbal algorithm show that the element concentrations in urine most frequently identified with eGFR, are Calcium (Ca), appearing 100%, Copper (Cu) and Zinc (Zn), appearing 96% and 80% of the time, respectively. These preliminary findings highlight the important interaction between Cu and Zn, potentially with Ca. The Cu/Zn ratio is a known critical marker of renal dysfunction, with imbalances potentially exacerbating inflammation, oxidative stress, and kidney damage. A stratified analysis examines the effect of gender and stage of CKD. Results indicate an association between CKD stage 1 and element concentrations in urine of Zinc (Zn) and Arsenic (As) (appear 84 and 68% in the compositional balance analyses). Further analysis will involve examining the association of As species, total As species and organic As species in the urine samples.

Keywords: Compositional balances, forward-selection method, renal disease, stratified analysis

Presenter: Jennifer McKinley, j.mckinley@qub.ac.uk

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Socioeconomic status and cognitive function in later life: examining 24-h time-use composition as a potential mediator

Aaron Miatke[1], Ashleigh Smith[1], Dot Dumuid[1], Ty Stanford[1], Jermon Drake[2], Audrey M. Collins[3], Maddison Mellow[1], Kelsey Sewell[3], Patricio Solis-Urra[3], Lauren Oberlin[3,4], Cristina Molina-Hidalgo[3], Peter Gianaros[2], Anna Marsland[2], George Grove[2] ,Haiqing Huang[3], Eric Vidoni[5], Amanda Szabo-Reed[5], Renee Rogers[5], John Jakicic[5], Jeffrey Burns[5], Charles H. Hillman[6], Arthur F. Kramer[6], Edward McAuley[7], Lu Wan[3], Chaeryon Kang[3], Kirk Erickson[3]

[1]Alliance for Research in Exercise, Nutrition and Activity, University of South Australia, Adelaide, Australia;
[2]Department of Psychology, University of Pittsburgh, Pittsburgh, PA, USA;
[3]Advent Health Research Institute, Department of Neuroscience, AdventHealth, Orlando, FL, USA;
[4]Department of Psychiatry, Weill Cornell Medicine, New York, NY, USA;
[5]Alzheimer's Disease Research Center, University of Kansas Medical Center, Kansas City, KS, USA;
[6]Center for Cognitive and Brain Health, Northeastern University, Boston, MA, USA;
[7]Beckman Institute, University of Illinois, Urbana, IL, USA.

This study demonstrates the application of compositional mediation analysis within a structural equation modelling (SEM) framework to investigate whether 24-hour time-use composition mediates the relationship between socioeconomic status (SES) and cognitive function in older adults. Data were from 589 cognitively unimpaired adults (mean age 69.8 years; 70.5% female) from the IGNITE study. Time use was measured via accelerometry and classified into sleep, sedentary behaviour (SB), light physical activity (LPA), and moderate-to-vigorous physical activity (MVPA). An isometric log-ratio transformation was applied using a sequential binary partition such that ilr1 represented the pivot coordinate for MVPA (ilr1: MVPA vs. remaining behaviours; ilr2: LPA vs. sleep+SB; ilr3: SB vs. sleep). Three SES measures were examined: area-level deprivation, subjective SES, and a latent objective SES indicator. Five cognitive domains served as outcomes: executive function/attentional control, episodic memory, processing speed, visuospatial function, and working memory. The SEM specified the three ilr coordinates as parallel mediators, and estimated paths from SES to each ilr coordinate (a paths), from each coordinate to each outcome (b paths), and direct paths from SES to outcomes (c' paths). Indirect effects were computed as products of a and b coefficients, with inference based on bias-corrected bootstrap confidence intervals. All outcomes were modelled simultaneously, allowing for correlated residuals. Age, sex, site, and race/ethnicity were included as covariates with standardised effects reports. Higher objective SES was associated with better cognitive performance across all domains (standardised total effects, i.e., sum of direct $'$ and indirect $a \times b$ paths: $\beta = 0.14-0.26$). Significant indirect effects (a×b products) through time-use composition, driven by ilr1, were observed for processing speed, working memory, and executive function/attentional control (standardised indirect effects: $\beta = 0.015 - 0.024$). Area-level SES showed similar indirect effects, while subjective SES demonstrated weaker, non-significant indirect pathways. This study illustrates how compositional mediation within SEM can accommodate compositional time-use data as mediators of multiple correlated outcomes.

Keywords: time use; mediation analysis

Presenter: Aaron Miatke aaron.miatke@adelaide.edu.au

Testing mean densities with an application to climate change in Vietnam

Camille Mondon[1], Huong Thi Trinh[2],
Josep Antoni Martín-Fernández[3], Christine Thomas-Agnan[1]

[1]Toulouse School of Economics, University of Toulouse Capitole, Toulouse, France;
[2]Faculty of Mathematical Economics, Thuongmai University, Hanoi, Vietnam;
[3]Universitat de Girona, Girona, Spain.

Given samples of density functions on an interval (a,b) of the real line, categorised according to a factor variable, we aim to test the equality of their mean functions both overall and across the groups defined by the factor. We build upon techniques from compositional data analysis (CoDA) but also from the Functional Analysis of Variance (FANOVA) framework. While the FANOVA methodology is well-established for functional data, its adaptation to density functions (DANOVA) is necessary due to their inherent constraints of positivity and unit integral. To accommodate these constraints, we naturally use the Bayes spaces methodology by mapping the densities with the centred log-ratio transformation into the space of square integrable functions on (a,b), where we can use FANOVA techniques. Many traditional contrasts in FANOVA rely on squared differences. When applied to the centred log-ratio transformation of the densities, these contrasts can be reinterpreted as squared distances between Bayes perturbations within the densities space. We compare five different test statistics. We illustrate our methodology on a dataset comprising a series of daily maximum temperatures across Vietnamese provinces between 1987 and 2016. For a given year and province, the series is treated as a sample from the hypothesised density of that province in that given year. The series are then smoothed into compositional splines densities. For each province, the time evolution is summarised by the slope of a density-on-scalar regression model where the scalar is time, thus defining the province's temporal trend in density space. Within the context of climate change, we first investigate the existence of a non-zero temporal trend of the densities of daily maximum temperature over Vietnam and then examine whether there is any regional effect on these trends. Finally, we explore interpretations based on infinitesimal odds-ratios, allowing to describe the trend's evolution more locally. Using the properties of infinitesimal odds ratios, we demonstrate that trend curves can be used to identify intervals where the probability mass has increased or decreased over time.

Keywords: analysis of variance, density data, functional data, odds ratio

Presenter: Christine Thomas-Agnan, christine.thomas@tse-fr.eu

Detecting outliers in compositional data using the logratio t model

Gianna Serafina Monti[1], Glòria Mateu-Figueras[2], Vera Pawlowsky-Glahn[2], Juan José Egozcue[3]

[1]Department of Economics, Management and Statistics, University of Milano–Bicocca;
[2]Department of Computer Science and Applied Mathematics, University of Girona;
[3]Department of Civil and Environmental Engineering, Technical University of Catalonia.

The orthogonal logratio Student's t distribution is a flexible probabilistic model for compositional data that generalises the logratio normal model. This distribution exhibits heavy tails, enabling improved detection of outliers. By defining the parameters of the distribution in orthonormal coordinates on the simplex, the Aitchison geometry is preserved.

In the context of robust compositional data analysis, several approaches have been proposed to mitigate the influence of atypical observations. Methods such as robust principal component analysis and robust covariance estimators, including the Minimum Covariance Determinant, are commonly employed to detect outliers by focusing on the central structure of the data distribution. However, these methods often rely on hard rejection rules or dimensionality reduction that may overlook the structural nature of heavy-tailed distributions.

We present an application of the logratio Student's t distribution for outlier detection, illustrating its ability to identify extreme compositions that deviate from the main structure of the data while avoiding misclassification of naturally dispersed points. By leveraging an intrinsic distance metric consistent with the simplex geometry, the logratio t model ensures invariance under logratio transformations and captures the covariance structure of compositional datasets in a robust manner. To further illustrate its performance, we use both a real data set and a simulation study. We compare it with the Atypicality index, a relative measure of atypicality of a composition formulated in terms of standard incomplete beta functions and assuming a Logistic Normal distribution as the base model. The obtained results confirm that our model consistently outperforms the logratio normal model and other robust estimators.

Keywords: orthonormal coordinates, Aitchison measure, multivariate t distribution, logratio normal family

Presenter: Glòria Mateu-Figueras, gloria.mateu@udg.edu

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Why nutrition data are compositional

Juliano Morimoto[1,2],

[1]University of Aberdeen, UK;
[2]Federal University of Paraná, Brazil.

Nutrition data is inherently compositional: nutrient intakes represent parts of a whole and therefore reside in the simplex rather than in real Euclidean space. Nevertheless, much of the nutrition and obesity literature continues to analyse these variables as if they were unconstrained, assuming distributions in R^n instead of the simplex S^n. This mismatch leads to statistical inferences that may be biased or conceptually inconsistent. In this contribution, I illustrate these issues using one of the most influential contemporary theories of obesity—the Protein Leverage Hypothesis (PLH). The PLH proposes that humans (and other animals) regulate energy intake to achieve a target level of protein consumption, thereby over-consuming fats and carbohydrates when the diet is protein-dilute. While the qualitative logic of the hypothesis remains intact, its quantitative formulation typically relies on statistical models that disregard the compositional nature of dietary data. By reframing nutrient intakes within a compositional data analysis (CoDa) framework, I show how the foundational relationships proposed by the PLH can be examined more appropriately. This approach clarifies the structure of nutrient trade-offs, avoids spurious correlations, and provides a more coherent basis for estimating the strength of "protein leverage". This work aims to highlight the relevance of CoDa methods for nutrition science, stimulate discussion on best practices for analysing compositional dietary data, and connect with experts interested in developing robust frameworks for applying CoDa in nutritional epidemiology and metabolism research.

Keywords: nutrition, geometry

Presenter: Juliano Morimoto, juliano.morimoto@abdn.ac.uk

Visualizing symmetric and skew-symmetric interactions in square compositional tables

Keita Nakamura[1]

[1]Tokyo University of Science, Japan.

Square tables, in which the same set of categories appears as both rows and columns, arise in a variety of contexts, including social mobility and repeated categorical measurements. When the analyst's primary interest lies in the relative structure of transitions rather than absolute frequencies, such tables are naturally treated within a compositional framework. In this setting, two fundamental challenges emerge: the diagonal entries tend to dominate the table and obscure the off-diagonal structure of primary interest, and the relative nature of the data necessitates analysis within the Aitchison geometric framework. To address these challenges simultaneously, we develop a weighted log-ratio analysis method for square compositional tables. Our approach employs an orthogonal decomposition of a square compositional table into symmetric and skew-symmetric components, each further decomposed into independence and interaction parts, thereby isolating diagonal influence while respecting the compositional structure of the data. A weighting scheme based on average arithmetic marginals is incorporated to stabilize the analysis against the high variability inherent in small components. Application of singular value decomposition to these weighted components yields separate visualizations of symmetric and skew-symmetric interactions. The proposed method is illustrated through application to a real dataset, highlighting the distinct roles of symmetric and skew-symmetric interactions.

Keywords: compositional tables, Aitchison geometry, orthogonal decomposition, log-ratio analysis

Presenter: Keita Nakamura, 6324703@ed.tus.ac.jp

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Who bears the loss? A compositional look at disaster payouts in Catalonia

Maria Isabel Ortego[1], Núria Pantaleoni Reluy[1], Núria Quesada[2], Nieves Lantada[1]

[1]Department of Civil and Environmental Engineering, Universitat Politècnica de Catalunya, Barcelona, Spain.
[2]Universitat Politècnica de Catalunya, Barcelona, Spain.

A fully functioning private insurance market for post-disaster recovery does not currently exist anywhere in the world, to the authors' knowledge. Governments have therefore stepped in to make disaster coverage available and affordable. Approaches vary widely, ranging from fully public programs funded and managed by the state to collaborative models like public-private partnerships that combine government involvement with private-sector participation. In Spain, insured losses from extreme natural disaster events are covered by the Consorcio de Compensación de Seguros (CCS), a public business organisation operating as an institutionalised public-private partnership. Relying on the publicly available CCS compensation data, this study analyses municipality-level payouts issued between 1996 and 2022 for damages resulting from natural disasters in the Catalonia region. The events examined include floods, storm surges, and extraordinary wind and rainfall. The dataset spans multiple assets, allowing a detailed assessment of the distribution of losses across social, economic, and infrastructure systems. As the sectoral compensation shares constitute compositional data, the analysis is conducted within the framework of compositional data analysis (CODA), ensuring coherence with the structure of data. Our main objective is to examine the relative shares of total compensation allocated to different sectors across municipalities and events using compositional data analysis. Within this framework, these relative compensation shares provide insights into sectoral vulnerability, reflecting the relative susceptibility of each type of asset to damage under hazardous conditions. By integrating spatial information through GIS and statistical tools, this approach supports the assessment of how frequency, intensity, and interaction of natural hazards translate into systemic socio-economic impacts at the municipal scale. The results may help to better understand the effects of these natural hazards. This evidence will be an essential support in order to identify priorities and make decisions for disaster preparedness and resilience in multi-hazard environments.

Keywords: insurances, natural disasters, relative compensation shares, vulnerability

Presenter: Maria Isabel Ortego, ma.isabel.ortego@upc.edu

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A trajectory-based clustering approach to intra-week time-use compositions

Miguel Angel Oviedo Caro[1], Ty Stanford[2], Aaron Miatke[2], Timothy Olds[2], Ty Ferguson[2], Rachel Curtis[2], Francois Fraysse[2], Adrian Esterman[2], Wendy Brown[2], Carol Maher[2], Dot Dumuid[2]

[1]University of Seville, Spain;
[2]Adelaide University, South Australia, Australia.

Time-use behaviours vary across the week in response to social, occupational and environmental constraints. Collapsing daily time-use into a single "average" weekly composition may therefore obscure important intra-week heterogeneity. We demonstrate how variation can be characterised using KmL3D, a multivariate longitudinal extension of the k-means clustering algorithm. KmL3D represents individuals as multi-dimensional trajectories across repeated time points and variables, and clusters them by minimising within-cluster distances between observed trajectories and cluster-specific mean trajectories. Distances are computed across all variables and time points using a Euclidean metric, with cluster centres updated iteratively until convergence. Multiple random initialisations are used to enhance robustness, and cluster quality indices guide selection of the optimal solution. We applied KmL3D to time-use data from 353 Australian adults, comprising daily minutes in sleep, sedentary time, light physical activity (LPA), and moderate-to-vigorous physical activity (MVPA). Thirteen months of daily observations were averaged for each day of the week, yielding seven time-use compositions for each participant. Prior to clustering, time-use compositions were expressed as isometric log-ratios and scaled (zero-phase component analysis whitening). Using 30 random initialisations and the Calinski-Harabasz index, a three-cluster solution was selected. Cluster 1 ($n = 138$) had the highest MVPA, particularly on weekends, at the expense of sleep rather than sedentary time. Cluster 2 ($n = 131$) had consistently low sedentary time and the highest LPA, especially on weekends. Cluster 3 ($n = 84$) had low physical activity (MVPA and LPA) throughout the week, with high sedentary time, like Cluster 1. Regression analyses indicated that, compared with Clusters 1 and 3, individuals in Cluster 2 had significantly lower body mass index and depression scores, and higher quality of life and happiness. By capturing joint temporal and compositional structure, KmL3D offers an alternative to average-based analyses, enabling the identification of behaviourally distinct and clinically relevant patterns across the week.

Keywords: longitudinal, clustering, time use

Presenter: Dot Dumuid, dot.dumuid@adelaide.edu.au

Analysing distributional and density data using the Bayes space framework in R

Ivana Pavlů[1], Raimon Tolosana-Delgado[2], Matthias Templ[3], Karl Gerald van den Boogaart[2]

[1]Palacký University Olomouc, Czech Republic;
[2]Helmholtz Institute Freiberg for Resource Technology, Germany;
[3]Fachhochschule Nordwestschweiz, Switzerland.

In this contribution, we introduce a newly developed R package designed as a practically oriented tool for the analysis of density and distributional data within the Bayes space framework. The package aims to facilitate the dissemination and practical adoption of recent methodological advances in density data analysis. We provide an overview of the key concepts underlying the package, its overall structure, and its main functionalities. Particular attention is devoted to the different types of input density and distributional data that users may wish to analyse, along with the corresponding data handling and preprocessing options. Finally, several illustrative examples are presented to demonstrate typical workflows and to promote the use of the package among a broader applied audience.

Keywords: density data analysis, software implementation

Presenter: Ivana Pavlů, ivana.pavlu@upol.cz

Test for the individuation of a common subspace in compositional datasets with structural zeros

Francesco Porro[1], Fabio Rapallo[1], Sara Sommariva[1]

[1] University of Genova, Italy.

In many real-world applications involving compositional data, the presence of structural (or essential) zeros is a common issue. Structural zeros occur when, due to intrinsic physical or biological constraints, one or more components of a composition are necessarily zero for a subset of the population under study. The classical Aitchison framework for compositional data analysis requires strictly positive components, as it relies on log-ratio transformations, which cannot be performed in presence of zero values. Consequently, datasets with structural zeros are typically partitioned into two subsets: one with the observations containing structural zeros and one with those without. Then they are analyzed separately, assuming the two datasets are drawn from two different subpopulations. However, this approach may lead to unsatisfactory results when the partition into two populations is merely artificial and due to the presence of zeros only. To address this issue, we propose a statistical test to assess whether the first K principal components of the two datasets generate the same vector space. Under the assumption of normality on the simplex, an analytical approximation of the null distribution is derived; otherwise, a nonparametric bootstrap approach is employed. Simulation studies show that the proposed test can effectively distinguish between scenarios where the two subsets share a common subspace and those where they are actually distinct. The method is further illustrated through an application on an experimental dataset concerning microbiome measurements.

Keywords: common subspace, logratio transformations, nonparametric hypothesis testing, principal component analysis

Presenter: Francesco Porro, francesco.porro@unige.it

Animal and plant protein intake and all-cause mortality: A 24-year compositional analysis in a population-based cohort

Federica Prinelli[1], Antonio Giampiero Russo[2]

[1]Institute of Biomedical Technologies - National Research Council, Segrate (MI), Italy.
[2]Epidemiology Unit, Agency for Health Protection of the Metropolitan City of Milan, Italy.

Background: The relationship between dietary protein intake and mortality, taking the source of the protein into account, has not been widely studied. Methodological challenges arise because macronutrients components are parts of a whole, making it diffucult to disintangle their specific effects from total energy intake. Furthermore, there is limited evidence on potential sex-specific associations, particularly in relation to unhealthy lifestyles or adverse metabolic profiles.
Objective: To investigate the association between dietary animal and plant protein intake and all-cause mortality in middle-aged adults over 24 years, using compositional data analysis (CoDa), while assessing effect modification by sex, lifestyle and metabolic risk factors.
Methods: This longitudinal, population-based study included 1,350 community-dwelling individuals (672 men and 678 women) aged 40–74 years. The participants were recruited between 1991 and 1995, and their mortality was monitored via a regional registry until 2015. Diet at baseline was assessed using a food frequency questionnaire, and animal and plant proteins were analysed using a CoDa approach based on the additive log-ratio (ALR) transformation. Cox proportional hazards models (adjusted for education, civil status, occupation, smoking, physical activity, alcohol consumption, fibre intake and energy intake) were used to estimate the association between protein intake and mortality. Effect modification by sex, smoking and obesity was tested via stratum-specific estimates. Results: Of the 405 deaths recorded, a higher intake of animal protein relative to starch was found to be associated with an increased overall mortality rate (hazard ratio (HR) 1.37, 95% confidence interval (CI) 1.00–1.87), particularly among males (HR 1.57, 95% CI 1.05–2.33). The strongest effects were observed among ever-smokers (males: HR 2.06, 95% CI 1.32–3.20; females: HR 1.90, 95% CI 1.18–3.06) and normal-weight participants (HR 1.64, 95% CI 1.03–2.59). In females, a higher relative plant protein intake was associated with lower mortality in lean individuals (HR 0.13, 95% CI 0.02–0.94), but with higher mortality in females with overweight/obesity (HR 5.81, 95% CI 1.52–22.17).
Conclusions: A dietary composition characterized by a higher relative contribution of animal protein is associated with increased all-cause mortality, particularly among males and among current and former smokers. In contrast, a higher relative intake of plant protein appears protective in lean females but is linked to a greater risk of mortality in females with overweight or obesity. These findings highlight the importance of considering dietary composition and support the need for tailored dietary recommendations that account for lifestyle and metabolic profiles.

Keywords: protein intake, all-cause mortality, compositional data analysis, population-based cohort study

Presenter: Federica Prinelli, federica.prinelli@cnr.it

A compositional approach to delineate fertile from barren veins in a gold mine

Sara Rezaei1[1], Hossein Molayemat[1], Farhad Mohammad Torab[1]

[1]Department of Mining and Metallurgical Engineering, Yazd University, Yazd, Iran.

Gold, as one of the most valuable commodities, is a hard target in all mineral exploration stages. The main reasons are the large nugget effect and the strong fluctuation of this element. One way to distinguish gold-bearing veins is the measurement of gold concentration in numerous large samples, which is, as expected, costly and time-consuming. Another approach is a statistical study of gold pathfinder multi-element geochemical data, which is known to be typically compositional. The sample space of geochemical compositions is the simplex. No compositional part can vary independently of the others. This is the source of spurious covariances and correlations. The present contribution aims at the discrimination of gold-bearing veins from barren ones in the Betze-Post gold deposit, Nevada. Four classes of gold mineralisation can be defined in the Betze-Post: (a) high-grade ore, (b) low-grade ore, (c) mineralised wall rock, and (d) waste. By considering the gold pathfinder element concentrations as input values, several classification methods are performed: (a) statistical methods such as Linear Discriminant Analysis (LDA) and cluster analysis; (b) machine learning methods such as K-Nearest Neighbors (K-NN), Support Vector Machine (SVM) and Random Forests (RF); (c) artificial intelligence methods such as shallow, deep and convolutional Neural Networks (CNN). Results show that using a set of logratios considerably increases the accuracy of gold vein classification. In our example, the compositional approach achieved the correct classification rate of 94%, and eliminated the ore/waste misclassification to 0. This is vital in mining affairs. None of the computational methods delivered such high rates of accuracy when the data is closed. The full separation of gold ore from waste is only seen after performing compositional data analysis. It is concluded that, despite the approach used, all the classification methods need a proper compositional representation before running.

Keywords: gold mineralisation, classification, logratio, machine learning

Presenter: Sara Rezaei, sararezaei@stu.yazd.ac.ir

A latent class approach for dynamic clustering of compositional data

Lorena Ricciotti[1], Sabrina Giordano[1], Antonello Maruotti[2]

[1]Università della Calabria, Dipartimento di Economia, Statistica e Finanza "Giovanni Anania", Cosenza, Italy;
[2]Khalifa University, Department of Public Health and Epidemiology, Abu Dhabi, UAE.

We propose a Hidden Markov Model (HMM) framework tailored for compositional data, which enables the identification of latent regimes and the study of their temporal evolution. Classification and modelling of such dependent time series or longitudinal compositional data can be challenging, as standard methods fail to account for the relative nature of the observations and the dynamic structure over time. To properly handle the simplex structure of compositional data, category-level expenditures can be treated in two alternative ways. In one approach, compositions are transformed using log-ratio techniques, mapping them to an unconstrained space where a flexible emission distribution, such as a contaminated normal, can be specified to capture variability within each latent state. Alternatively, the Dirichlet distribution can be applied directly to the untransformed compositions, providing a natural probabilistic model on the simplex. In this work, we compare these two approaches in terms of their ability to capture latent and temporal dynamics, highlighting the strengths and limitations of each. Parameter estimation is carried out via a specialised Expectation-Maximisation (EM) algorithm adapted for compositional HMMs, which jointly estimates state-specific Dirichlet parameters, transition probabilities, and the effects of covariates. Covariates can influence either the emission distributions, the state transitions, or both, enabling the analysis of how socio-economic and demographic factors affect both the composition of household food expenditure and the probability of switching between latent consumption regimes. The approach proposed is applied to data from the 2021 Italian Household Budget Survey (HBS), focusing on a six-part composition of household food expenditure: bread and cereals, meat, fish, cheese, fruit and vegetables, and beverages. Compositional vectors are obtained by closing category-level expenditures to the total household food budget. The model successfully identifies latent consumption regimes, captures their dynamic transitions over time, and quantifies the impact of household characteristics on expenditure patterns. Overall, this framework provides a flexible and interpretable method for the classification and analysis of longitudinal compositional data. It is broadly applicable to other domains involving dynamic relative measurements, offering insights into both persistent and transient patterns in dependent compositional time series.

Keywords: hidden Markov models, compositional time series, latent variables, model-based clustering

Presenter: Lorena Ricciotti lorena.ricciotti@unical.it

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A Bayesian support vector machine approach for compositional data classification

Irene Ríos Martín[1], Irene Creus Martí[1], Francisco José Santonja[1]

[1]Department of Statistics and Operational Research, Universitat de València, Spain.

In their 2011 and 2017 studies, Polson, Wenzel, and collaborators introduced a Bayesian framework equivalent to the Support Vector Machine (SVM), which enables probabilistic estimation of model parameters and quantification of predictive uncertainty. In this work, we extend this Bayesian framework to compositional data, explicitly accounting for the relative structure of the components and adapting the model to the specific characteristics of multivariate compositional datasets. To demonstrate the effectiveness of the proposed approach, we apply it to cross-sectional microbiota datasets and evaluate its performance in comparison with several commonly used classification methods.

Keywords: microbiome classification methods, Bayesian compositional data analysis

Presenter: Irene Ríos Martín, irene.rios@uv.es

Some thougths on L1-type penalisation norms in convex clustering for compositional data

Jordi Saperas-Riera[1], Paula de la Lama Zubiran[1], Josep Antoni Martín-Fernández[1]

[1]University of Girona, Spain.

In a previous edition of this conference (CoDaWork24), we introduced a convex clustering framework for compositional data based on the Clusterpath methodology, highlighting the use of a compositional variable representation and proposing the L1-CoDa norm as a penalisation term. In the present contribution, we extend this line of research by investigating new L1-type norms and their theoretical implications within convex clustering for compositional data. Specifically, we study the L1-clr and L1-plr norms, and analyse their properties in comparison with the previously proposed L1-CoDa norm. We examine how these norms influence the clustering path, the fusion of clusters, and the geometric structure of the solutions. Our analysis reveals that the choice of norm plays a fundamental role in shaping the clustering process, leading to distinct behaviours and interpretative properties. The presentation places particular emphasis on the theoretical aspects of these norms, providing insight into their mathematical structure and their suitability for compositional data analysis. To illustrate the practical consequences of our theoretical findings, we include an application to a real dataset.

Keywords: convex clustering, clusterpath, L1-norm

Presenter: Jordi Saperas-Riera, jordi.saperas@udg.edu

Calibrating centered log-ratio biplots

Phuti Sebatjane[1], Sugnet Lubbe[2], Niel le Roux[2]

[1]University of South Africa, South Africa;
[2]Stellenbosch University, South Africa.

Biplots are powerful graphical displays for multivariate data in a Euclidean sample space that simultaneously display the rows/samples (as points) and columns/variables (as vectors or calibrated axes) of a data matrix, revealing important characteristics or patterns in the data. Constructing biplots for compositional data requires an appropriate coordinate transformation, such as the centred log-ratio (clr) transformation, before applying ordinary principal component analysis. The biplot shows the samples as points and the parts (or variables) as vectors or biplot axes. However, visualising the parts needs special consideration: in addition to visualising the parts as vectors or calibrated axes, all log-ratios are visualised as so-called links. In this paper, we explore different ways to represent the links. In particular, we enhance the compositional data biplot by extending the calibration of biplot axes as proposed by Gower & Hand (1996) to the links. An illustration of how to calibrate both the biplot axes and the link vectors is given, subsequently representing the compositional data biplot with calibrated axes (biplot axes or link vectors). Different units for the calibrations can be computed, e.g. clr units, the original parts or, preferably, the log-ratios. In addition, we show how the calibrated axes can be translated away from the centre of the data without changing the inner product representation of the biplot. A dataset of the fatty acid composition of dairy milk is used as an example to show the usefulness of the compositional data biplot and our extensions. For example, similarities between individuals (dairy cows) across different seasons and correlation patterns between groups of parts (saturated, monounsaturated and polyunsaturated fatty acids). We conclude by highlighting properties that the compositional data biplot with calibrated axes or link vectors shares with the scatter plot, e.g., calibrated axes and orthogonal projection of row points to read off approximated values of the samples.

Keywords: centred logratio coefficients, compositional biplots, fatty acids, principal component analysis

Presenter: Phuti Sebatjane, sebatpn@unisa.ac.za

Functional future of health science

Paulíina Ševčíková[1], Aleš Gába[1], Dorothea Dumuid[2],
Željko Pedišić[3], Jitka Machalová[1], Jana Pelclová[1], Karel Hron[1]

[1]Palacký University Olomouc, Czech Republic;
[2]University of South Australia, Australia;
[3]The University of Hong Kong, China.

Movement behaviours such as physical activity, sedentary behaviour, and sleep are interrelated components of daily life and should not be analysed in isolation. As parts of a movement behaviour composition, they are intrinsically constrained and require statistical approaches that account for their compositional nature. Traditional analyses rely on discretising raw accelerometer signals into predefined movement behaviour categories; however, such discretisation inevitably introduces an artificial structure that may obscure meaningful patterns present in the original data. This contribution proposes an alternative approach to analyse movement behaviours without loss of important information: instead of working solely with conventional movement behaviour classifications, the underlying acceleration signal across the day is represented as functional data. Characterising the original intensity distributions as smooth functions preserves the continuous nature of accelerometer-derived data and avoids premature aggregation into discrete categories. Within this approach, statistical modelling is carried out using function-on-scalar regression, in which the response is a functional object (daily acceleration curves) and the predictors are scalar quantities derived from compositional data. More specifically, we employ scalar explanatory variables given by pairwise logratios constructed within backward pivot coordinate systems of the movement behaviour composition, ensuring scale invariance and interpretability in the compositional context. To further analyse the functional regression results and assess their implications for behaviour change, we perform a compositional isotemporal substitution analysis. The methodology will be illustrated using the cross-sectional study among Czech school-aged children, demonstrating its practical relevance for time-use epidemiology.

Keywords: time-use data, functional data analysis, function-on-scalar regression.

Presenter: Paulína Ševčíková paulina.jaskova@upol.cz

Ecological succession of bacterial communities on single sand grains

Sebastián Silva-Solar[1], Rudolf Amann[1], Katrin Knittel[1]

[1]Max Planck Institute for Marine Microbiology.

Understanding the ecological principles that govern community assembly remains a central challenge in biology. Microbial systems have emerged as powerful models for studying these principles, offering highly diverse communities at tractable spatial and temporal scales. Here, we track how diverse marine microbial communities from subtidal sediments colonised sterile sand grains using a controlled microcosm experiment. We show the ecological succession of these bacterial communities at the scale of single sand grains, and describe their diversity and dynamics over time using 16S rRNA amplicon sequences. We observed remarkably similar successional patterns at the genus level between sand grains from the same and different bottles. Mainly, the dynamic was led by Vibrionaceae at the early stage, Rhodobacteraceae and Alteromonadaceae at the mid stage, and Flavobacteriaceae and Saprospiraceae at the final stage. Overall, communities on individual sand grains converge over time while communities from the liquid fraction do not. Furthermore, by pooling the communities from several sand grains, we observed around 10000 different ASVs per bottle per time, and concluded that more sampling effort is required to reach a saturation point. Our results highlight ecological consistency despite the astonishing diversity of these communities, with strong selection and recognisable taxonomic patterns emerging during the colonisation of free surfaces in sediments.

Keywords: microbial colonization, metabarcoding, community assembly, 16S rRNA

Presenter: Sebastián Silva-Solar, ssilva@mpi-bremen.de

Spline approximation of probability density functions

Stanislav Škorňa[1], Jitka Machalová[1],

[1]Palacký University Olomouc.

Approximation of probability density functions plays a key role in statistics, data analysis, and applied mathematics, particularly when the underlying distribution is unknown or only indirectly observed. Among other approaches, spline-based methods provide a flexible and mathematically well-founded framework for constructing smooth density estimates while retaining local adaptivity and computational efficiency. This contribution presents a general overview of density approximation using spline functions. We focus on representing probability density functions as linear combinations of spline bases, such as B-splines and, in particular, ZB-splines. The focus will be on applying functional data analysis to real data and on methods.

Keywords: probability density function, spline approximation, density data analysis

Presenter: Stanislav Škorňa stanislav.skorna01@upol.cz

Isotopologue distributions are compositions

Alberte Sloth Carlsen[1], Te Chen[1], Teddy Groves[1], Nicholas Luke Cowie[1], Christian Brinch[2], Lars Keld Nielsen[1,3]

[1]Novo Nordisk Foundation Biotechnology Research Institute for the Green Transition (BRIGHT), Technical University of Denmark, 2800 Kongens Lyngby, Denmark,
[2]DTU National Food Institute, Technical University of Denmark, 2800 Kongens Lyngby, Denmark,
[3]Australian Institute for Bioengineering and Nanotechnology, The University of Queensland, Brisbane QLD 4072 Australia.

Quantifying metabolic fluxes is crucial for understanding cellular metabolism. The internal metabolic flux cannot be directly measured because internal metabolite concentrations are subject to homeostatic regulation. Isotope-based Metabolic Flux Analysis (I-MFA) is a standard approach to estimate intracellular metabolic fluxes. This technique infers fluxes by comparing simulated and measured isotopologue distributions (IDs) of metabolites from isotope labeling experiments. An isotopologue is composed of the same molecule with a difference in the number of neutrons. Because IDs represent relative fractional abundances that strictly sum to one for given metabolites, they are inherently compositional data. However, state-of-the-art estimation approaches rely on calculating standard Euclidean distances between IDs in a non-compositional paradigm.

Our study proposes a new compositional I-MFA method. We demonstrate how to construct a meaningful orthonormal basis for IDs via ordered sequential binary partitioning, which can be used to perform isometric log-ratio (ILR) transformation. As a minimal change to existing I-MFA workflows, we suggest estimating fluxes by minimising Euclidean distances between ILR-transformed IDs.

We compared our method with the state-of-the-art method using a well-known toy example model and a larger, biologically realistic, model. In each example, a set of simulated IDs was generated based on a simple but structurally plausible measurement model, which were used to estimate the metabolic fluxes. We performed sensitivity analysis by varying the true fluxes used as inputs for the simulation, and used Monte Carlo simulation to compute confidence intervals. The result showed that our proposed method tended to make more accurate flux estimates with narrower confidence intervals compared to the state-of-the-art estimation approaches. We conclude that compositional data analysis is necessary for I-MFA, and potentially relevant to other fields in biology that deal with relative abundances.

Keywords: isotope, metabolism, metabolic flux analysis, ILR

Presenter: Te Chen, techen@dtu.dk

Personalised prediction of cognitive outcomes and traffic lights to signal positive changes in time-use composition

Ty Stanford[1], Maddison Mellow[1], Bethany Robins[1], Ashleigh E Smith[1], Dot Dumuid[1]

[1]Adelaide University, Australia.

The relationship between behavioural time-use composition (TuC) and outcome variables (e.g., quality of life, cognition, health outcomes) is often interrogated within the compositional data analysis (CoDA) framework using isotemporal substitution. Isotemporal substitution demonstrates modelled average change in the outcome from TuC reallocations relative to a reference TuC (often a compositional mean or median). One-for-one or one-to-proportional-remaining isotemporal substitution has traditionally provided TuC reallocation effects along each compositional part individually (e.g., sleep or physical activity) but not in tandem. However, exploration of TuC reallocation effects can be extended to be more focused and purposeful if the TuC reallocations are depicted as the proportionate (standardised) distance travelled in the direction (or opposite direction) of the optimal TuC. To do so, we require: (a) a model estimated optimal TuC with respect to the outcome, and (b) a vector projection method to calculate the proportionate travel, in a compositional variance standardised space, from the reference TuC towards or away from the optimal TuC. We name this method: Best EstimAted Compositional-Optimal Navigation (BEACON). Framing reallocations this way allows both personalised reference TuCs ("current") and end-user self-determined acceptable TuC changes ("proposed") to be dynamically calculated and visualised over a single axis as the proposed proportionate travel along the BEACON path from the current to optimal TuCs. In addition to outlining the calculations required for the BEACON method, we demonstrate how BEACON was used to help determine personalised aspirational TuC changes to improve cognition in an Australian cohort of older adults, by way of a traffic light system, within the Small Steps study, which aimed to improve activity and sleep habits to decrease dementia risk.

Keywords: optimal composition, vector projection, isotemporal substitution, behavioural time-use

Presenter: Ty Stanford, tyman.stanford@adelaide.edu.au

First application of CoDA to the high-dimensional (HD) biological data matrices of single-cell RNA-sequencing (CoDA-hd R package)

Nelson Tang[1], Jinghan Huang[1]

[1]Department of Chemical Pathology, Li Ka Shing Institute of Health Sciences, Faculty of Medicine, The Chinese University of Hong Kong, Hong Kong SAR.

CoDA has been applied to high-dimensional (HD) biological data matrices like gene expression and microbiome, which are typically of the size 1000's samples x 10,000's. Single-cell RNA-sequencing (scRNA-seq) produces new HD gene-expression data matrices with extremely sparse data, at 100,000 cells x 20,000 genes. In scRNA-seq, each biological cell is a sample, and the expression levels of all 20,000 genes are quantified. Although scRNA-seq is inherently compositional, broader adoption of CoDA has been limited by the abundant "dropout" zeros, which make it incompatible with CoDA. Therefore, the common practice is to scale data and apply standard Euclidean-based methods for analysis. We argue that CoDA is the most parsimonious statistical method for analysing scRNA-seq. The first major contribution is a scalable strategy to apply CoDA to matrices with thousands to millions of cells. Gene expression counts can be transformed into log-ratios (LR), such as the centred-log-ratio (CLR), which maps the data into Euclidean space while retaining some CoDA features: scale invariance and permutation invariance. Standard log-ratio analysis (LRA) becomes computationally prohibitive in this setting, so we employ a fast, truncated Singular Value Decomposition (SVD). This "CLR + partial SVD" pipeline yields a close approximation to full LRA but runs in minutes to hours, making CoDA-hd practically useful for routine large-scale scRNA-seq studies. The second key innovation is a new treatment of zero counts via a new count-addition scheme. Conventional CoDA typically adds a fixed constant to all entries, which performs poorly on highly sparse scRNA-seq data. We instead develop column-specific addition methods (different addition values for each sample based on the total count of the sample). SGM (sum of individual sample/ geometric mean of all samples sum) method, where the numerator is the sum of counts per cell (sample) and the denominator is the geometric mean of total counts across all cells. Using variable numerators and denominators preserves cell-specific structure while reducing the inflated-zero effect. We also show that commonly used log-normalised matrices can be converted into CoDA log-ratio representations, enabling CoDA-hd analyses on existing public datasets without requiring raw counts. CoDA-hd demonstrated improved performance in downstream analyses, including cell clustering and dimensionality reduction. An important contribution of CoDA-hd is greater robustness to zero-inflation, and it correctly clusters low-quality cells (samples with many biological zeros) away from other cell types to avoid misinterpretation. The method is implemented in the CoDA-hd R package, which offers the first CoDA-based analysis application for scRNA-seq datasets and is more robust than current practice based on Euclidean-space statistics.

Keywords: single-cell RNA-sequencing, CLR, SVD

Presenter: Nelson Tang, nelsontang@cuhk.edu.hk

A critical comparison of handling zeros in high dimensional compositional count data

Wenqi Tang[1], Kamila Fačevicová[2], Klaus Nordhausen[3], Sara Taskinen[3]

[1]University of Jyväskylä, Finland;
[2]Palacký University, Olomouc, Czek Republic; [3]University of Helsinki, Finland.

The growing use of high-throughput sequencing (HTS) has enabled the large-scale production of compositional count data, driving progress in microbiome research. However, such count data are often highdimensional, over-dispersed, and heavily zero-inflated, and they conflict with the continuity assumptions underlying log-ratio–based compositional data analysis (CoDA), creating substantial methodological challenges. This review provides a overview of zero-handling strategies in compositional data, covering zero-tolerant transformations, imputation approaches for rounded zeros, and statistical models for essential zeros. We specifically highlight the problems that arise when applying the logratio framework to sequencing-derived compositional count data, where violations of continuity can induce numerical instabilities and biased statistical inferences. Motivated by these issues, we systematically examine how existing imputation strategies behave when adapted to discrete, zero-inflated count data, including an evaluation of how the discrete, lattice-valued nature of the data affects imputation performance. Overall, this review consolidates scattered methodological developments, clarifies appropriate use cases, and identifies open challenges that motivate future zero-handling frameworks capable of jointly accommodating compositional constraints, zero inflation, and the lattice nature of count data, while also providing a detailed discussion of the comparison results.

Keywords: zeros, discrete compositions, Microbiome count data, imputation

Presenter: Wenqi Tang wenqi.t.tang@jyu.fi

Pairwise ratio-based differential abundance analysis of infant microbiome 16S sequencing data

Dennis E te Beest[1,2], Kevin Mildau[1,2], Bas Engel[1,2], Gerrit Gort[1,2], Jolanda Lambert[3], Sophie H N Swinkels[3], Fred A van Eeuwijk[1,2]

[1]Biometris, Wageningen University;
[2]Research, 6700 HB Wageningen, The Netherlands;
[3]Danone Nutricia Research, Uppsalalaan 12, 3584 CT Utrecht, The Netherlands.

Differential abundance analysis of infant 16S microbial sequencing data is complicated by its challenging data properties. These properties include high sparsity, extreme dispersion, and the relative nature of the information contained within the data. We propose to analyse these data with a pairwise ratio analysis using a beta-binomial regression model. The resulting method provides a flexible approach to analysing sequencing data and conducting differential abundance analysis that does not require zero imputation. The beta-binomial regression model itself is not new, and several readily available implementations exist. This means our method can be combined with techniques commonly found in the regression toolbox, such as random effects and splines. We also demonstrate how pairwise results can be used to select interesting taxa, without overinterpreting the pairwise result. Based on simulation results, we discuss some strengths and weaknesses compared to analysing pairwise log-ratios with linear regression.

Keywords: pairwise ratio, microbiome, beta-binomial model

Presenter: Dennis te Beest, dennis.tebeest@wur.nl

Subcompositions-based tree methods

Raimon Tolosana-Delgado[1,2]

[1]Helmholtz Institute Freiberg for Resource Technology, Helmholtz Zentrum Dresden-Rossendorf;
[2]Institute for Stochastics, Technical University Bergakademie Freiberg.

In CoDaWork 2019, we presented a review of the most common machine learning methods, discussing the possibility of adapting them for compositional data. Tree methods (classification and regression trees, random forests and the like) were particularly tricky to adapt to compositional data because their conventional implementation builds splitting rules only using one marginal of the covariables. If the covariables form a composition, then these rules amount to using the raw absolute percentage of each component. As a reminder: a classification tree is a hierarchical set of rules, where each branching rule is optimised to achieve a binary split of the data set which best increases the purity of the resulting data subsets, e.g. measured with the Gini index. A regression tree does the same by minimising the post-split pooled variance of the response. A random forest is a set of trees, each trained with its own bootstrap sample, where the branches are each time optimised over a subset of the available covariables.

This contribution presents a series of modified random forest algorithms to exploit compositional information as covariates. A new algorithm has been implemented, which uses on each branch a random subcomposition to find the optimal splitting criterion. The decision can be based on the new marginals (after closure) or on all possible pairwise logratios within this subcomposition. These are compared against the original algorithm with two case studies, but also with respect to their theoretical abilities (e.g. when dealing with values below the detection limit and true zeros).

Keywords: random forests, subcomposition

Presenter: Raimon Tolosana-Delgado, r.tolosana@hzdr.de

Isometric log-ratio transformations for interpreting enzyme-leach geochemical signatures in volcanic terranes: insights from Hispaniola arc systems

Ricardo A. Valls Álvarez[1]

[1]Valls Geoconsultant.

Interpreting soil geochemical data in deeply weathered tropical volcanic arcs remains a major exploration challenge due to low-contrast anomalies, strong lithological overprints, and the inherent compositional nature of multi-element datasets. Enzyme-leach soil geochemistry, while highly sensitive to subtle metal dispersion through thick regolith, produces data that are subject to closure effects, rendering traditional univariate thresholds, raw ratios, and standard multivariate techniques prone to spurious correlations and misleading interpretations. This contribution demonstrates how isometric log-ratio (ilr) transformations, embedded within a compositional data analysis (CoDa) framework, provide a mathematically robust and geologically meaningful solution to these challenges. The study is based on a comprehensive enzyme-leach soil survey from the La Majagua prospect, located in the Cretaceous–Paleogene Greater Antilles volcanic arc of the Dominican Republic. The dataset comprises 349 soil samples and 51 variables, including base metals (Cu, Zn, Pb), pathfinder elements (Co, Ni, V), Rare Earth Elements (REEs), and ancillary physical parameters such as pH, density, elevation, and selected geophysical responses. The geological setting includes meta-tonalite intrusions, amphibolitic and green schists, and extensive alluvial and eluvial cover, creating a complex geochemical environment where distinguishing hydrothermal signals from lithological background is critical. Initial inspection of raw data revealed extremely high correlations among REEs and strong associations between certain elements (e.g., Ni) and lithology, illustrating the dominance of closure-driven relationships. To overcome this, a Sequential Binary Partition (SBP) was constructed, informed by geological reasoning and exploratory AI-assisted correlation screening. The resulting ilr balances successfully separated soil material types (alluvial versus eluvial), lithological domains (tonalitic versus schistose), and discrete multi-element assemblages indicative of Cu–Zn–Pb $\pm$ Co–Ni–V mineralization. Ilr-derived maps revealed spatially coherent copper enrichments in southern alluvial zones, zinc–lead clusters associated with meta-tonalites, and several high-priority targets not detected using centred log-ratio PCA or percentile-based methods. Furthermore, ilr coordinates proved essential as a feature-engineering step for machine learning, ensuring that predictive models are trained on valid Euclidean variables rather than mathematically inconsistent raw concentrations. Overall, this work highlights ilr-based CoDa as a transferable, scale-invariant, and exploration-ready workflow for enzyme-leach surveys in regolith-dominated volcanic arcs worldwide.

Keywords: enzyme-leach soil geochemical data, machine learning, mineral exploration, isometric log-ratio transformation

Presenter: P. Geo. Ricardo A. Valls Álvarez, vallsvg@gmail.com

Proportionality between allocations in asset management

Juan David Vega Baquero[1], Miguel Santolino[1]

[1]University of Barcelona, Spain.

Asset allocation refers to deciding the optimal participation of each asset within a portfolio. Therefore, these participations are a composition, and compositional methods should be used to analyse them. When trying to find relationships between parts of a composition, proportions are more suitable than correlations. In this paper, using a previous proportionality index as starting point (the Proportionality Index of Parts, PIP), two new indices are proposed, and all of them are used to analyse the asset allocation in a portfolio composed of five stocks from the IBEX 35 (the Spanish stock market index). These allocations are obtained using the Minimum Variance method developed by Markowitz. The proposed indices focus on subcompositions formed by two components to assess the proportionality between them. On one side, using the properties of the logratio variance, it is possible to find its maximum and minimum values and define the Pairwise Proportionality of Logratios (PPL). On the other side, the variances of the components of the two-part subcomposition are assessed, obtaining their limits and defining the Pairwise Proportionality Index (PPI) accordingly. Results show low proportionalities between the allocations through the PIP and PPL, but higher ones for the PPI. However, all three indices show consistent values and shed light on the connection between the volatility of stocks, the allocation method and the proportionality of the allocations.

Keywords: Aitchison geometry, capital allocation, portfolio theory, proportionality

Presenter: Juan David Vega Baquero, jvegabaq38@alumnes.ub.edu

Using X-means and Imputation to Obtain Geologic Insights from Incomplete Inorganic Geochemical Data (XRF)

Matthew Wehner[1]

[1]Veratek Design, California, USA.

A major limitation of using portable X-ray fluorescence (pXRF or HHXF) in geologic applications is that they usually contain many missing data, which has stymied analysis by the usual classification methods like K-means. Also, any global statistics should honour the compositional nature of the data (CoDa). I show that these pXRF datasets still have significant utility and can provide insights, such as using a dataset from the Cretaceous-aged shales (informally known as the Eagle Ford Shale) of central and west Texas, USA. Specifically, I show that rock classification can be performed on such incomplete datasets without prior knowledge of the number of classes. I start with imputing the missing data using methods adapted for compositional data analysis (CoDa), such as the log-ratio SVD or log-ratio EM. Then I used an SVD approximation method and then truncated the number of components by calculating the optimal threshold that distinguishes a component that contains information above the Gaussian noise-level (instead of using heuristics like the elbow method or finding the number of components that account for $90+\%$ of variance). Then the lower rank matrix is recovered from the components above the calculated optimal threshold, denoising the dataset. Since the number of rock classes is not known a priori, I will avoid methods that require the number to be specified, and I chose to use the X-means method on each dataset, which determines both the number of classes and assigns each XRF analysis to a rock class. X-means is an extension of K-means in that it solves for an optimal number of clusters, starting with two clusters and increasing the number using the Bayesian Information Criterion (BIC); other criterion methods are possible. The goal is to allow the data to "speak" for itself regarding such questions as the bulk composition and its stratigraphic distribution, the number of practical rock classes based on inorganic geochemistry, regional changes of paleoredox states, and covariation of organic matter with other elements.

Keywords: geochemistry, unsupervised classification, denoising, XRF

Presenter: Matthew Wehner, nwgeologist@gmail.com

Moran's I for compositional data: A measure of spatial auto- and cross-correlations

Takahiro Yoshida[1], Ikuho Yamada[1].

[1]Center for Spatial Information Science, The University of Tokyo, Japan.

Moran's I statistic has been widely used as a fundamental tool for evaluating spatial autocorrelation in univariate data. While various extensions have been proposed to accommodate different data types—most notably the Generalised Moran's I statistic—methodological developments for compositional data remain insufficient. Compositional data, characterised as multivariate vectors whose components are strictly non-negative and sum to a constant, present unique challenges due to their inherent constraints. A rare exception is the approach based on canonical correlation. However, their methodology is limited to assessing only the overall degree of spatial autocorrelation across the entire multivariate set, failing to isolate the specific contributions of spatial autocorrelation from the spatial cross-correlations between different variables. To address these limitations, this study proposes a novel Moran's I statistic specifically designed for compositional data by incorporating the principles of compositional data analysis (CoDA). Our proposed framework allows for a comprehensive evaluation of (1) overall spatial autocorrelation, (2) individual spatial autocorrelation for each variable, and (3) individual spatial cross-correlations between different variables. For the empirical application of this proposed statistic, we used the building-uses data provided by the Tokyo Metropolitan Government. Building-uses were classified into three primary categories: commercial, industrial, and residential. A compositional dataset was then constructed by aggregating the total gross floor area of each category at the neighbourhood level (called cho-cho-moku). The analysis yielded an overall spatial autocorrelation coefficient of 0.64, suggesting that urban areas with similar building-use compositions tend to form significant spatial clusters. Furthermore, by decomposing the spatial auto- and cross-correlations, we identified a negative spatial cross-correlation of –0.38 between industrial and residential uses from the perspective of the industrial use. This result can be interpreted as a clear manifestation of urban planning policies, where land-use guidance through zoning regulations effectively separates industrial and residential uses. Finally, this study explores prospective research avenues, focusing on temporal dynamics of spatial correlations and comparative performance assessments.

Keywords: Spatial auto- and cross-correlation, Moran's I, building-uses, spatial statistic

Presenter: Takahiro Yoshida, yoshida@csis.u-tokyo.ac.jp